

[U] XDemics



Harvey Mudd College Engineering Clinic

Multi-Axis Motion and Vibration System for Bioreactor Cell Seeding and Recovery

Full-Year Report

Fall 2025 - Spring 2026

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Abstract

XDemics designs innovative bioreactors to increase cell culture throughput and expand access to life-saving biologics by improving the scale and efficiency of cell seeding and harvesting. Currently, a key bottleneck in the process is the manual manipulation (rotation, vibration, etc.) of cell culture containers, which prevents large-scale production and introduces user variability. The XDemics clinic team was tasked with designing, constructing, and testing a robotic system to automate bioreactor cell seeding and recovery in order to decrease the risk of contamination, increase throughput, and improve reproducibility. This paper begins with a literature review of hydrodynamic damage research and existing robotic and microfluidic systems to provide background on the scope and technical details of the assigned project. The Technical Approach section introduces the team's design process and the development of a fully integrated proof-of-concept automated system, including a structural frame to support all required bioreactor manipulations, an adjustable tilt plate for controlled wetting, adaptable fixturing for different bioreactor geometries, and an integrated vibration mechanism for cell detachment and resuspension. The Experimental Work section highlights that a minimum tilt angle of 15° is required to achieve complete membrane wetting without trapped air, ensuring uniform surface coverage for cell seeding, while the implemented vibration system achieved greater than 95% cell detachment and resuspension efficiency, confirming its effectiveness in facilitating cell recovery. These results validate the key functional components of the design and demonstrate that a simple, mechanically driven tilting, rotation, and vibration approach can reliably perform the necessary bioreactor operations. The final system represents a scalable proof-of-concept for automating HDCR operation, integrating mechanical, electrical, and biological considerations into a cohesive platform and establishing a foundation for future development of fully automated, high-throughput bioreactor systems.

Sponsors

XDemics is a startup company originated from California Institute of Technology focused on advancing cell growth technologies to transform biomedical research and development (R&D). Their patented high-density cell respiration (HDCR) technology overcomes the common issue of inefficient oxygen transfer during cell growth by using breathable silicone grooves. This allows for a high-density cell growth of mammalian cells reaching 10-100x higher than traditional methods like petri dishes and stirred tanks. The technology also reduces R&D and manufacturing costs and ensures quality, scalability, and efficiency for scientists and manufacturers. XDemics has its headquarters and manufacturing facility in San Rafael, CA, and biological testing laboratories in Novato, CA. For this clinic project, technical founder and CTO Colin Cook (PhD), senior engineer Gabriel Zwillinger (HMC '24), XDemics VP of Engineering Nicholas Scianmarello (PhD), and Director Core Technologies & Field Application Specialist Austin Santiago are the appointed liaisons who provide information, guidance, and opinions throughout the project.

Literature Review

Robotic System

An instinctive solution for a multiaxis robotic system is a robotic arm. The application of robotic arms in automating rotation and agitation is ubiquitous in a variety of assembly processes. One such paper investigated the use of the ABB IRB 140 [1] multiaxis robotic arm coupled with a vibrating disk to reduce friction in the mating process of automated assembly [2]. The literature proved the robot effective in attaching force/torque sensors and interacting with a separate vibration system. This research provides inspiration for the team's ideation process in considering the vibration/agitation and rotation components as separate systems that interact with one another through their attachment to the bioreactor.

The team found that the solution calls for a less complex system than the multi-jointed robotic arm. The solution operates in as few axes as possible to produce the rotational motions needed for cell seeding and harvesting. 2 axes of rotation are optimal. A study on a novel robotic bioreactor (see Figure 1) for in vitro tissue engineering provides inspiration for an adjustable attachment system and actuation system for linear motion [3]. The study 3D printed the robotic bioreactor, which is feasible to repurpose and produce with readily available materials at Harvey Mudd College.

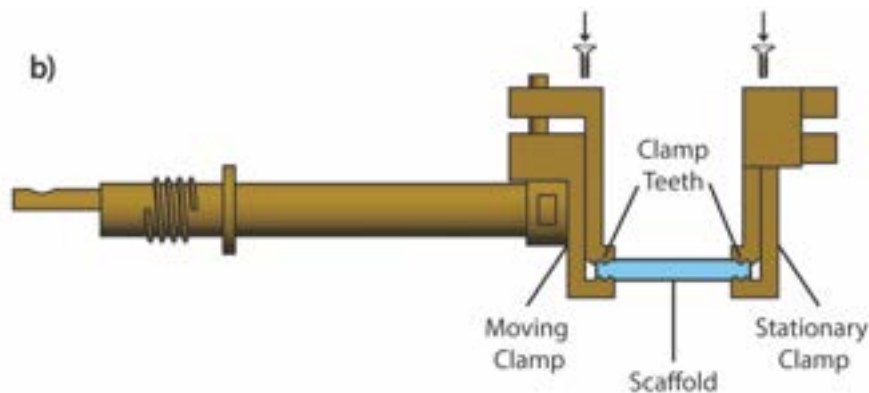


Figure 1: Schematic from “Design and Development of a Robotic Bioreactor for In Vitro Tissue Engineering” for a 3D printed robotic bioreactor using a clamp system to hold a scaffold in place. The design is included in this paper for inspiration on the robotic system’s attachment to the bioreactor. [3]

Applying the proposed system to this project, the clamp teeth could provide a fixed attachment to the top and bottom of the bioreactor so as to not disturb the determined compression of the stacked silicone sheets inside the platens. In relation to the tubes attached to the bioreactor, the scaffold would be where the bioreactor is placed with the clamps attached along the long axis of the rectangular prism. If a pneumatic piston is attached to the moving clamp where the long tube is in Figure 1, the clamp system would be pushed back and forth to agitate the bioreactor along the short axis. The productivity of cell throughput of this system depends on the optimal axis for agitation. Agitation along the long axis would require a different piston placement and more complicated attachment. For the team’s design, a rotating motor can also be attached to the moving clamp in order to rotate the system. The study’s robotic bioreactor

also has base dimensions that fit in most conventional incubators to account for the cells' environmental conditions, and similarly can be considered for the project.

By transitioning to an automated robotic system, the advantages and challenges will be met. According to the paper, "Automated manufacturing of cell therapies," by Alice Melocchi et al, automation provides the benefits of repeatability, accuracy of tasks, improved management of operations, utilizing sensing and controls, and reducing the need for human involvement [4]. It also allows for a smaller footprint and multiple simultaneous operations to support scalability. However, automation would reduce process visibility, which can be important to operators, and it would not completely eliminate failures. There are also regulatory considerations when applying an automated system to biotherapeutic products.

Microfluidics System

In order to automate cell seeding and harvesting, the manipulation of the bioreactor heavily relies on producing the desired microfluidic patterns to properly seed and remove cells from the bioreactor's silicone grooves. In the academic paper, "Engineering challenges in high density cell culture systems," Sadettin S. Ozturk highlights the limitations of working with high cell culture densities in engineering the reactor, the fluid medium, and the mixing process. [5]. The paper discusses the need for higher exchange rates, volumetric flow rates, and the challenges with balancing this need with increases in susceptibility to clogging and loss of separation efficiency of the cells. The article also mentions a decrease in mixing efficiency due to an increase in viscosity for high cell concentrations which leads to increased mixing times. These limitations are important in understanding the effect of higher cell culture densities on the manipulation process and components of the robotic system. If the system is more susceptible to clogging, then the agitation function of the system might need to be repeated between steps and/or be able to agitate along multiple axes to dislodge the cells from the bioreactor's wells.

Furthermore, this article emphasizes the importance of achieving proper mixing in order to promote homogeneity in the cell culture. One of the objectives for this project is uniform cell seeding and controlled cell agitation, of which vibration is an effective means. However, challenges arise when determining the proper method and intensity of vibration without

damaging the cells. The robotic system should also be easily operable and functional over a long period of time to accommodate higher cell culture densities. To test the microfluidics for the automated robotic system, controlled parameters can be set, such as the cell density, peristaltic pump flow rates, and culture parameters of humidity and temperature [5].

Hydrodynamic Damage to Animal Cells

This project requires special attention to the impact of hydrodynamic forces on live cells. Yusuf Chisti wrote about this in his article, “Hydrodynamic Damage to Animal Cells,” for the Journal of Critical Review in Biotechnology. Dr Chisti details how animal cells are damaged by the hydrodynamic forces applied to the cells during culture processes. There are many sources of hydrodynamic stress. For example, bioreactors are mechanically stirred vessels that create varying shear rates and turbulence. Process equipment like transfer pipes, valves, filters, and turbulent jets can generate localized high rates of shear force. Turbulence causes isotropic turbulence that can impose stresses that are dependent on energy dissipation. [6]

Shear forces are particularly important to the process of automating cell culture as many bioreactor manipulations create shear stress. Shear is expressed as shear stress or shear rate for which calculations are given to estimating stress from different equipment such as cell culture vessels, bioreactors, transfer piping, and recovery operations. Components such as impeller tips and capillary entrances create much higher stresses than bulk averages. These different hydrodynamic forces have cellular responses. Mechanical rupture can occur when membrane tension exceeds a critical threshold causing the cell to burst. Physiological effects of sub-lethal shear can impair growth, metabolism, or trigger apoptosis — a type of programmed cell death where cells die without damaging neighboring cells. Extensional flow at constrictions or nozzles can be particularly damaging to cells. [6]

Cell sensitivity to these processes varies drastically by cell type and growth phase. Factors that influence damage are cell size, age, membrane properties, media compositions, duration and frequency of exposure to high shear stress. Rupture forces tend to be an order of magnitude higher than the sub-lethal forces. One example is in the Hybridomas process that fuses b cells with myeloma cells to create hybridoma cells for immunizing an animal with

antigens. In the Hybridomas process for a turbulent viscometer , “under turbulent conditions, damage occurs when shear stress exceeds 5 Pa” [6, p. 87]. Dr. Alongkorn Pimpin’s paper investigates the effects of shear and extensional stresses on cells in a spiral microchannel and contraction-expansion arrays shown in Table 1 [7]. However, shear stress has a power law relationship. If the stress is increased by 10x, the cells have a 100x shorter exposure time for survival.

Cell Type	Effects of Shear and Extensional Stresses on Cells
Red Blood Cells (RBCs)	Abnormal morphology above 150 Pa, rupture at 300-400 Pa
White blood Cells (WBCs)	Functional changes at 15 Pa for 10 minutes
T cells	Tolerate up to 20 Pa for 10 min
Hybridoma Cells	Viability loss begins at 0.33 Pa, and 50% loss at 78 Pa for 30 minutes or 19 Pa for 2 hours
Insect Cells	50% loss at 1.5 Pa after 3.5 hours
HEK & CHO	Apoptosis at 1.09-3.05 Pa
RSC96 & L8 CELls	Damage at 100-1000 Pa for 10-120s
Endothelial cells	Phenotypic changes at 0.4-0.8 Pa for 15-16 hours

Table 1: Effects of Shear and Extensional Stresses on different Cell types [7] The left most column describes types of cells and the right column contains the shear stress in Pa and the impact on the cells.

Technical Approach

The technical approach for the project follows the general guidelines of the design process, beginning with thorough scoping of the problem and identification of key objectives, constraints, and required system functions. From this foundation, the team progressed through iterative stages of research, ideation, evaluation, and prototyping to develop and refine potential solutions. Design alternatives were systematically assessed and converged into a final proof-of-concept system, which was then constructed and experimentally validated. This structured approach ensured that the final design not only met the functional requirements of the bioreactor system, but was also grounded in both engineering principles and experimental results.

Project Delineation

To begin, the specifications of the project were broken down into objectives, functions, and constraints. In *Engineering Design*, objectives are defined as the “feature or behavior that we wish the design to have or exhibit”, constraints as “a limit or restriction on the features or behaviors of the design” that cannot be violated in a successful design, and functions as “things a designed device or system is supposed to do” [12]. Organizing the project in this way makes sure that all user specifications are met and allows for thorough prototype evaluation later on.

Objectives

- Be compact
- Be reusable
 - Reuse between cycles
 - Easily cleaned between cycles
- Be low particle generating
- Be lightweight
- Be automated, excluding setup and cleanup
 - Completely replacing technician work, minus setup and cleanup
- Be easy to use for non-engineers
- Be low-cost
- Be efficient

- Find the best angle, flow rate, optimal movements, etc
- Be consistent in cell throughput and viability
- Be adjustable to the given range of bioreactor sizes
- Be easy to scale up for future iterations with larger sizes of bioreactors
- Be robust
- Be able to produce the same cell throughput and linear scalability as manual manipulation
- Be able to modify the plate-handling procedure

Constraints

- Must support cell seeding, transfection, and harvesting
- Must not cause damage to plates/wells (i.e., through use of hydrostatic pressure)
- Must support 60 lbs of weight
- Must support a 8" W x 18" L x 18" H block
- Must operate in a closed system
- Must accommodate the tubes attached to the plate
- Must operate under laboratory humidity and heat (100% humidity, 37°C)
- Must be compatible with solvents and chemicals used in a clean room

Specifications/Functions

- Automate cell seeding
 - Rotate bioreactor along short axis
- Automate cell harvesting
 - Invert and agitate bioreactor
- Automate transfection/fluidizing the cells for virus loading
- Securely attach to bioreactor
- Remain in set position during failure or power outage

Task Breakdown

Task 1: Project Management Plan and Kickoff

Task 1.1 Project Kickoff and Preliminary Liaison Meetings

The project began with an in-person meeting between the clinic team, the clinic team advisor, and the clinic liaisons with the purpose of introducing the team and getting to know the project. During the meeting, we scheduled a weekly meeting time with the liaisons and clarified their communication and project management expectations. In the first two weeks, weekly liaison meetings and the first clinic design review involved clarifying the desired outcomes of the project by redefining the problem statement and solidifying the project's deliverables, objectives, constraints, and specifications. Early consensus on the project's goals helped to direct the focus of the clinic team in the ideation and evaluation of design alternatives and eventual convergence onto a single prototype.

Task 1.2 Project Management

The team co-leads are Erin Wang and Leilani Elkaslasy in the Fall, and Andrew Williams and Joshua Heinstein in the Spring. They are seniors in the clinic team and are tasked with managing and directing the project including:

- Scheduling internal meetings, advisor meetings, and liaison meetings, including hosting virtual meetings and planning meeting agendas
- Organizing site visits, including purchasing flight tickets and arranging transportation
- Submitting Clinic deliverables in the internal Clinic team folder, including the Team Charter, Work Plan, Budget, and Midyear Report, Final Report, Project Days Presentation and Poster
- Overseeing major communications between the team advisor and the clinic liaisons
- Outlining the project timeline through the Work Breakdown Structure (WBS) and Gantt Chart
- Hosting ideation boards that are accessible to the team and the liaisons
- Checking one-on-one with the team members to gather individuals' desired goals and feedback for the Clinic semester
- Updating the direction and dynamics of the team based on given feedback

All deliverables were presented to the team advisor and liaisons before submission.

Task 2: Research and Initial Ideation

Task 2.1.1 Research

Once the project is fully scoped, the team conducted a literature review of current methods of cell seeding and harvesting and applications of microfluidic theory in increasing cell throughput, as well as existing robotic systems with multi-axis and/or agitational functionality. Additionally, a subteam uses COMSOL to simulate the feasibility of using gas induced vibrations to agitate the cells.

Task 2.1.2 Site Visits

The team used the Fall Semester site visit to fully realize the constraints and functions of the robotic system by gathering documentation of the manual bioreactor manipulation and bioreactor manufacturing process, sharing initial ideations with low resolution models as deemed helpful for clear communication, and gaining feedback on the capabilities of the proposed alternatives. The Spring Semester site visit was used to test the final prototype, specifically the percent of cells that could be dislodged from their wells by mechanical agitation.

Task 2.2.1 Initial Ideation

There are four main technical areas that this project covers upon as outlined in the project description: mechanical system design, microfluidics, and cell biology:

1. Mechanical Design: The ideation for the robotic system mainly involves the mechanical design of the robot and was split into three main subsystems: rotation, fixturing, and agitation. All CAD models are made on the team's Fusion 360 shared folder.
2. Microfluidics: The microfluidic portion of the design involves the flow of fluids in and out of the system. The team focused on investigating the feasibility of fluidic induced vibrations to ignite the cells, and concluded it was not realistically feasible. All necessary fluid simulations were run on COMSOL.

3. Cell Biology: For the Fall Semester, the team used microspheres to mimic mammalian cells for experimentation. The team plans to continue using microspheres for testing a higher-resolution prototype in the Spring. The team tested a finalized prototype in the Spring on mammalian cells instead of microspheres to verify the cell throughput.

Task 2.2.2 Documenting Ideation with Functions

The first session of ideation was held at an internal meeting where the produced designs were assigned to randomized subteams for further research and ideation. The subteams' research was shared on an online brainstorming whiteboard platform, like Lucid. The subteams provided greater flexibility in scheduling ideation time and promoted collaboration and connection between individuals on the team. The advantages, disadvantages, and other findings of each alternative were discussed in the following internal meeting.

In order to keep track of and continue expanding on possible means for each function, the team utilized an FMEA chart that organizes each function with its means. The team also kept descriptions and visual models of the design alternatives in the team's shared drive.

Task 3: Evaluating and Converging Design Alternatives

Task 3.1 Evaluation of Initial Ideations

By this point, the team had gathered an extensive list of possible design alternatives that encapsulate the desired functions. These design alternatives were evaluated on their ability to meet the constraints and achieve the desired objectives in order of importance using a Best of Class chart and/or testing low-resolution prototypes. The team also used the Design History File template provided by the liaisons to continuously document the justifications behind design selection.

Task 3.2.1 Converging on A Design

Once the design alternatives have been evaluated and ranked, the best design(s) were selected to move forward into a round of rapid prototyping to verify that the mechanisms are feasible and functional. Given verification, the team proceeded to the prototyping phase.

Task 3.2.2 Prototyping the Chosen Design

Prototyping began with CAD and 3D-printed mechanisms that proved out the design at hand. The design was evaluated with each new prototype until it has been improved and polished. Then, after ordering the necessary parts, a fully functional prototype was made. However, a control system was not implemented, because the team decided to focus on aspects of the project that were more crucial to a proof of concept.

Task 3.2.3 Evaluating the Chosen Design

The chosen design was evaluated at the Spring Semester site visit. The evaluation is outlined later in this report. The team tested and validated the effectiveness of the agitation system, which met the given criteria from XDemics.

Project Deliverables

Fall Deliverables

- Manually test and replicate bioreactor manipulation to automate.
- Design and prototype a low-resolution robotic system(s) to manipulate the bioreactor with a variable motion profile.
- Converge on the best design to pursue in the spring.
- Write the Midyear report to explain and justify our design decisions, and provide a testing plan for optimizing the motion profile.

Spring Deliverables

- Test and optimize the motion profiles for cell seeding and recovery.
- Build a mechanical prototype of a robotic system that can rotate and agitate cells.
- A plan/list of materials for scaling the size of the system up.

- Write the final report, which explains our final product and provide test results from automated cell seeding and recovery.

User Design Requirements ([link](#))

Critical user Requirement (see to name)	Design Inputs (Specifications)	Design Outputs (Sprints, drawings, BOMs, procedures)	Tools	Results	Owner
Must be able to manipulate profusion reactor	Device supports full movement envelope for a 50 stack tall device for a wet and dry mass, including sterility, fluid mass. Justification: bioreactor will need to be manipulated with varying amounts of liquid and masses inside the stack at well	Mechanical drawings for motion profile, torque/speed specifications, load rating for motors and attachments between subsystems	Attach various masses at different positions on the bioreactor and operate through full range of motion. Verify system stability and control.	NA, testing in progress	Rotation subsystem/Full using subsystem
Must support wetting	System removes ~ 90 percent of bubbles during wetting/agitation phase. Justification: Remaining bubbles inside the silicon grooves will prevent cells from settling into the grooves and greatly decrease cell throughput	Wetting sequence procedure, Tilt angle specifications for wetting	Visual inspection, bubbles before vs. after wetting sequence ~ 90% reduction	NA, testing in progress	Wetting subsystem
Must support failures (e.g. power outage, system glitch)	Automated safe state and resume procedure. System locks in place when event occurs and logs the failure. Justification: A loss of power could trigger electrical components to fail and result in loss of stability and control	Motor safety and locking specifications, Load rating of motor without power, software/firmware for logging event and system behavior	Conduct 10 power-loss test at different stages of operation, verify safe shutdown and correct lock and resume behavior	NA, testing in progress	Rotation subsystem/Full or subsystem
Usable while wearing standard PPE (e.g. no gloves)	No screws or small tools required for handling system in the clean lab. Handles and latches operable with full gloves 1 size larger than optimal fit. Justification: In the clean lab, there cannot be any small components and users are wearing gloves which reduces handling dexterity	Assembly drawings and CAD, mechanism design, ergonomic spec	Conduct 10 test runs of full operation procedure with users wearing clean lab-gloves or vinyl gloves that are 1 size larger than their optimal size, measure success rate and time, verify that steps requiring interaction with system run smoothly	NA, testing in progress	Full system/Full subsystem
Must dislodge cells from walls effectively	System supports upside-down rotation and agitation in the upside-down position at a defined amplitude/frequency. Justification: An upside-down position inserts the walls to allow for cells to move away and out from the silicon grooves. Agitation will likely be necessary to completely dislodge cells from walls	Agitation profile, motor load and control specifications, bioreactor measurements for rotation clearance	Conduct 10 test runs of cell agitation. After agitation inspect for remaining cells, quantify cell detachment, verify system has not failed nor deformed and can continue multiple agitation cycles without wear	NA, testing in progress	Agitation and experimental on subsystem/Rotation subsystem
Must not damage cells	Shear stress on cells of applied to cells < TBS. Maximum agitation and rotation acceleration < TBS. Justification: Quick or harsh movements	Motion profile spec, computational fluid simulation or CFD/DOE, manual procedure	Conduct 10 test runs of full operation procedure. Before and after each agitation and each rotation, visual inspection of cell viability	NA, testing in progress	Fluids subsystem/Full system

Table 2: User Design Requirements (see Appendix A for full size version) sheet details the critical requirements of the proposed system for the user, the design inputs the requirements provide, the design outputs to be considered when ideating alternatives for the system, testing for meeting the requirements, results of the tests, and the subteams that have ownership over meeting the given requirement.

The User Design Requirements (see Appendix A for full size version) were requested as a documentation deliverable by the liaisons. The document was used to identify the requirements of the robotic system for a biologist in a clean room, to ideate what parts of the system will need to be designed around which requirements, and to plan testing for measuring whether the system meets the user requirements. It is to be noted that testing will occur with the final, full-resolution prototype, and thus, the results column has not been filled out for the low-resolution prototype drafted in the Fall semester. The full document can be accessed through [this link](#).

Failure Mode and Effects Analysis (FMEA)

Process Step / Item	Potential Failure Mode	Potential Effects of Failure	Severity (S)	Potential Causes of Failure	Occurrence (O)	Current Controls	Detection (D)	RPN (S×O×D)	Recommended Actions
(Describe the step or item being analyzed)	(What could go wrong?)	(What happens if failure occurs?)	(1–10)	(Why it could happen)	(1–10)	(Existing preventive/detective controls)	(1–10)	(Auto-calculated)	(Suggested mitigation actions)
Pin	Pin could break inside the pinhole	Possibly means bioreactor is stuck to fixture and need to replace whole system	10	Choose the wrong pin, or pin experiences unforeseen amount of force	1	Buy a pin rated for calculated forces or above	2	20	Choose a pin rated above the forces expected
	Pin might fall out	Bioreactor falls out of fixture and cells are now everywhere :(	10	Mechanism holding pin failed	3	Spring loaded ball that is default locked, human detection	8	240	Possibly a stop on the system that only allows the system to move if the pin is locked; add a second pin (one pin on each side); choose pin that mitigates this failure mode
	Pin might get loose	Not strongly coupled so vibrations and force transmissions are weaker	9	Vibrations might cause pin to shake loose	3	spring loaded ball, human detection	3	81	Buy vibration resistant pin, has stronger springs and more balls.
	Pinhole might break	Bioreactor falls out of fixture, or bioreactor can't be removed.	10	Combination of vibrations and forces during operation might wear the pinhole down during use.	4	Human detection	2	80	Make the pinhole out of something strong that won't break or deform like stainless steel.
	Pin getting jammed in pinhole	Can't close or open fixture. Can't start or stop process.	8	Friction? Things falling into pin hole	5	Human detection.	2	80	Make the pinhole go all the way through the part so that anything that falls in can go out the other side.
	Pin gets lost	Can't secure bioreactor.	6	Pins aren't captive	4	Lead rope tying it to the robot.	4		Make pins captive, or connected to machine by a wire. Use a latch instead?
Latch to secure fixture to bioreactor	Latch slides open during vibration	Bioreactor falls out of fixture and cells are now everywhere :(	10	Latch improperly closed. Fatigue. Latch mechanism fails, becomes more likely after the below failure	3	Human detection.	2	60	Tight-Hold draw latch uses springs to mitigate vibrations
	Latch breaks	Possibly means bioreactor is stuck to fixture and need to replace whole system	8	Choose the wrong latch, or pin experiences unforeseen amount of force	1	Human detection	2	16	Place latch such that it experiences minimum amount of forces acting against it.
Placing bioreactor into the fixture	Bioreactor gets stuck midway and can't be taken out (entrance)	Whole system needs to be replaced	8	Deformation of fixture due to temperature or stress	1	Human detection	5	40	Manufacturing the fixtures with the right tolerance grades.
	Bioreactor gets stuck midway and can be removed	Whole system needs to be replaced	7	Deformation due to temperature or stress	1	Human detection	5	35	Manufacturing the fixtures with the right tolerance grades.
	Bioreactor expands and starts binding to the fixture	Pins can pop out, bioreactor can get stuck	10	High temperatures	2	human detection	2	40	Is this a flaw with the bioreactor or the robot? Maybe potentially make the fixture adjustable size?
Tubing Clip	Clip Break	Tube tangle	6	Tubes too tight creates torque	1	Strong Clip- Slack in Tube	8	48	Clear user instructions- Testing failure scenarios to better design improvements
	Tube Pinch	Restrict Tube Flow	9	Torque- overtighten clamp	3	Too little slack in tube length.		0	
	Generate Particles							0	
Tilt mechanism	Motor stops working	Tilt is fixed at certain angle	4	Lose power, motor use failure	1	Motor safemode	8	32	Integrate safemode with some alert/detection system
	Block get stuck in wedge	Tilt is fixed at certain angle and can't be removed	5	Misalignment of wedges and block, weight of bioreactor/system too heavy	1	Human detection	2	10	Ensure sturdy materials, track system to keep block in place, account for many times expected weight
	Supports fall	Entire bioreactor system falls	10	Too much weight, stress over time	1	Human detection	2	20	Ensure system can support much more than expected weight

Table 3: Preliminary FMEA - see appendix B for a full sized version

The purpose of the Failure Mode Effects Analysis study is to think through potential points of failure or malfunction in order to identify the key areas of improvement and understand the risks associated with design alternatives. The team learned how to conduct an FMEA this semester and plans to conduct more in depth FMEA for all aspects of the design throughout the year.

Design Alternatives

The following section describes the design alternatives created, expanded, and considered during the ideation process. The section starts with a morphological chart to introduce the initial means generated for each of the project's objectives. Then, the alternative designs for rotating the bioreactor are compared, with the spinner design resulting as the most effective and simple

design to prototype. Based on the spinner design, the section reports the design alternatives for stopping the motor during a system failure, fixturing the bioreactor to the rotation mechanism, managing the tubing on the bioreactor, and vibrating the bioreactor.

Morphological Chart

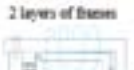
Functions	Means		
Rotate bioreactor along the short axis	 Tilt Plate	 Face rod and motor	 Gyroscope rings
Tilt bioreactor along the long axis	 Motor spinner	 Rail system	 Gyroscope rings
Invert bioreactor (not limited to along the short axis)	 Motor spinner	 Face rod and motor (short axis)	 Gyroscope rings (short axis)
Agitate bioreactor	 ERM coin vibration	 Scotch yoke	 Microfluidic Pulses
Attach to bioreactor	 Clamps	 Free-spinning fixture	 2 layers of frames
Remain in set position during failure	 Control system Lock	 Battery Power	 Wheel Brakes
Adjust to multiple bioreactor sizes	 Dovetail	 Slider	 Lunchbox

Table 4: The Morphological Chart (see Appendix C for full size version) lists the objectives of the project and multiple methods of reaching each objective individually. The chart was used to identify designs that met multiple objectives, to visually compare the complexity of design

alternatives for one objective, and to connect possible design alternatives that could be combined into one overarching system.

To manage the initial ideation process, a morphological chart (see Appendix C for full size version) was made. For each of the functions of the design, several means were generated. Each mean defines a way in which that function can be performed. For rotating, tilting, and inverting the bioreactor, similar means were generated, with most involving using a motor to rotate the bioreaction. These means turned into the complete-system prototypes: the Gyroscope, Flip Flop, and Spinner. These are detailed later in the design alternatives section.

Rotation

To determine the most effective mechanism for controlling the bioreactor's orientation during cell seeding, resting, wetting, and harvesting, several design alternatives were developed and evaluated. Each concept aimed to balance the need for precise rotational control with practical considerations such as complexity, manufacturability, tubing management, and system reliability. The following section presents four proposed mechanisms—the Spinner, Gyro-Bot, Flip-Flop, and Tilt Plate outlining the operational principles, advantages, and limitations of each. This comparison informed the selection of a primary prototype and guided subsequent design refinements to meet the functional requirements of the bioreactor system. Ultimately, the spinner was chosen to move forward with the full prototype that was built during Spring Semester.

Alternative: Gyro-Bot

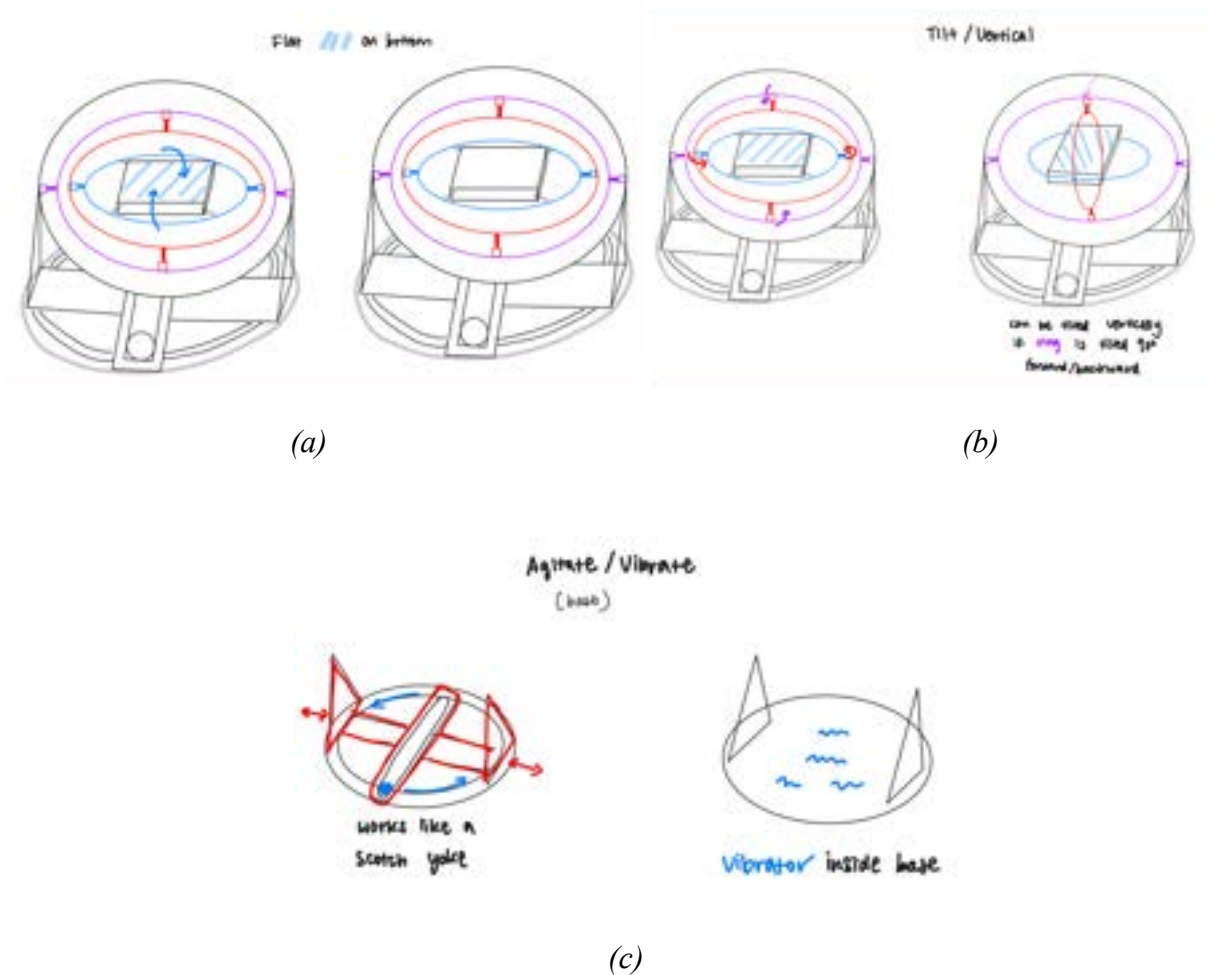


Fig. 2: The Gyro-Bot is able to (a) rotate around the axis of flow completely upside down using the innermost ring (blue) and (b) tilt to over 90° along the axis perpendicular to flow using the second ring (red) and the third ring (purple). The outermost ring (black) attaches the ring system to the base, where the vibrating mechanism (c) is either a scotch yoke that turns rotational motion into single-axis motion or a vibrator located inside the base.

The Gyro-Bot mechanism was considered as a possible design due to the freedom of movement in every axis, and thus also the ability to agitate in every axis. The design is inspired by the gyroscope and the larger-scale Aero-trim. The first innermost ring (blue) rotates the bioreactor around the axis of flow (see Fig. 3a). The second ring (red) rotates the bioreactor around the vertical/z-axis that runs through the center of the bioreactor, which then allows the

third ring (purple) to rotate the bioreactor along the axis perpendicular to flow (see Fig. 3b). The outermost ring (black) holds the fixture to a base that either uses a scotch yoke mechanism (rotational motion to single-axis motion) or an inset vibrator for agitating the whole ring system (see Fig. 3c). This ring system allows for the bioreactor to be rotated and tilted at any degree in all directions. After evaluation, the Gyro-Bot was not chosen as a feasible prototype due to the major challenges of managing the tubes with the rotation motions and the complexity of the system that introduces more areas of failure, especially when adding agitation. Although the ability to move in multiple axes is attractive in our deliverable to optimize the cell seeding and harvesting process, we decided that the Gyro-Bot actually provided superfluous axes of rotation, whereas other design alternatives met the necessary axes required with simpler designs.

Alternative: Flip-Flop

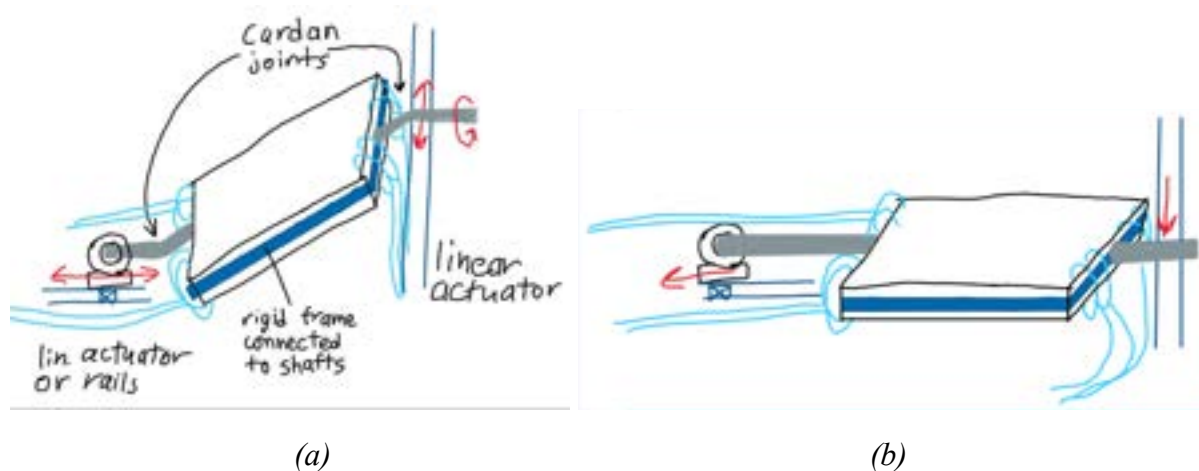


Fig 3: The Flip Flop mechanism is depicted in 2 configurations. In Figure 3a the bioreactor is rotated 45 degrees around the short axis, with the Cardan joints bent. The red arrows point in the directions of the moment along the two linear actuator systems. In Figure 3b the rotation is reversed by movement along the linear actuators.

The Flip-Flop mechanism was considered in order to rotate the bioreactor in two axes. It was designed based on the assumption that the bioreactor would require two axes of rotation, while the Spinner and Gyro-Bot were designed for one and three axes of rotation. The Flip-Flop would allow for 180 degrees rotation around the long axis, and 45 degrees rotation around the short axis. The long-axis rotation is driven by two motors, each with a shaft fixed to the

bioreactor. The short-axis rotation slides the bioreactor back, and the force is driven by the linear actuator motion. The use of Cardan joints allows the rigid frame to pivot and maintain its connection to the actuators even as its angle changes. The angle of this rotation is limited to the maximum bend of the Cardan joints. The highest maximum operating angle for Cardan joints on McMaster Carr is 45 degrees [8]. The shafts would be connected to a frame that wrapped around the sides of the bioreactor.

The complexity of the flip-flop had the potential to pose more failure points than other design options, with the many joints and axles involved. The fixture shaft also had the potential to interfere with the tubing, offering exposed rotation points for entanglement. Furthermore, the flip-flop was bulk, with the separation of two spread-out components.

The flip flop mechanism inspired using a simple axel based design for rotation, realizing that only the long axis needed to rotate 180 degrees and the short axis could rotate to a smaller extent inspiring the team to decouple the two axes

Spinner

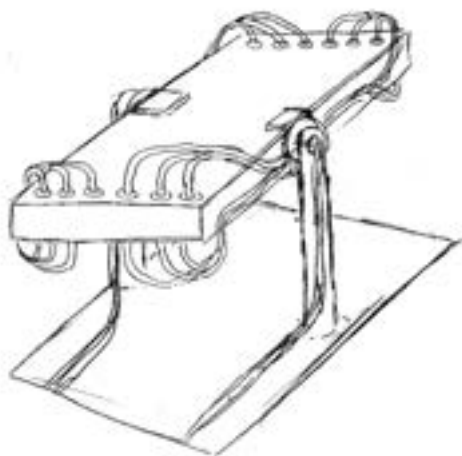


Fig. 4: Drawing of spinner design including tubing management for mitigating tubing stretching and pinching. The design incorporates an axle that the bioreactor is attached to and rotates around. The axle is fixtured to a plate for stability and attachment to a rigid body.

The spinner design mounts the bioreactor on a stand with two built-in motors. The motors pivot the bioreactor around one axis up to 180 degrees. This allows the device to seed the cells at a 90-degree angle, rest during growth at some slight angle, and finally turn 180 degrees to release the cells from the silicone grooves.

This design is favorable because it has only one axis of rotation and only 1 or 2 motors. The spinner's simplicity makes it a good first prototype because having a functional low-resolution prototype will also help the team test for optimization so that the design can later be refined. This project has a lot of different options and requires a lot of testing for optimization, so the sooner we have a working prototype, the better.

One main concern was that the tubes might restrict motion, but splitting the tubes down the middle and running them along the side reduces the extra slack length needed and keeps them from tangling. This was especially useful because many of the other designs required a lot of slack to move the bioreactor around without straining the tubes.

Next steps would probably focus on optimizing angles for cell seeding and resting. We would also test different methods of wetting the membrane and releasing the cells since this design lacks a vibration or agitation mechanism. Finally, we would need to look at attachment mechanisms to distribute the strain over the entire bioreactor.

Final Design: Spinner



Fig 5: CAD rendering of the spinner assembly, showing the support frame, motor-driven belt system, fixtures, and mounted bioreactor used to rotate and position the bioreactor during wetting and cell-processing operations.

The final assembly was designed to provide a reliable and manufacturable system capable of supporting bioreactor wetting, agitation, and rotational positioning. A simplified single-axis spinner design was ultimately selected because it reduced mechanical complexity while still meeting all functional requirements for controlled bioreactor rotation. This approach improved reliability, simplified fabrication and assembly, and minimized the number of moving components compared to more complex multi-axis systems. To achieve the required wetting orientation, the spinner assembly was integrated with a hinge-based tilt plate mechanism, allowing the bioreactor to reach controlled tilt angles while maintaining structural stability.

Several design choices were made during prototype construction to improve performance and ease of operation. A belt-driven motor system was used to provide smooth and consistent rotational motion while isolating vibration from the motor. Custom fixtures with slotted features were incorporated to securely align and support the bioreactor during operation while still allowing easy insertion and removal. The frame was designed with a rigid, open structure to

improve accessibility to the system components and simplify testing and adjustments during development. Additional considerations, including component spacing and fixture tolerances, were incorporated to improve vibration transfer and ensure repeatable positioning throughout operation.

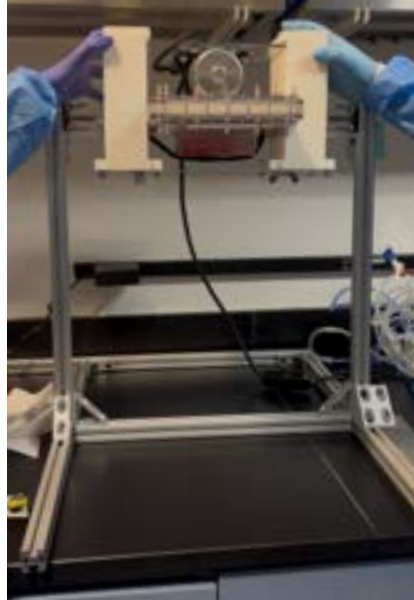


Fig 6: Physical full-scale prototype of the Spinner being tested at the site visit complete with fixture, vibrator couplings, free-spinning shafts, and an 8020 built frame

Tilt Plate

The tilt plate design would be coupled with the spinner to provide an extra axis of rotation for wetting. Pushing only one leg of the spinner up would complicate the symmetry of the motors and coupling of rotation to the bioreactor. Thus, the spinner would sit atop of the tilt plate such that the whole spinner mechanism would tilt rather than one leg lift higher than the other. Based on wetting tests, this mechanism would need to provide a minimum tilt of 15° .

Early Prototype



Fig 7: The early tilt plate design uses a horizontal lead screw to push the wedges of a plate such that the plate tilts along a single axis. The mechanism would be used in tandem with the Spinner design alternative to provide a second, more constrained axis of rotation for wetting. The manufacturability and necessity for delicate balance classifies this design as undesirable for the final tilt mechanism.

Due to the lack of ease in manufacturability, the early prototype tilt plate design is not recommended for use as the tilting mechanism. However, certain components and ideas were valuable from this prototype and made significant influence into the recommended design.

The early prototype design consisted of a top plate with symmetrical wedges at the bottom and a rod across the middle that can rotate (tilt) between two legs. The two legs are attached to a base plate for anchoring. A horizontal lead screw motor that moves along the middle of the base plate pushes against the wedges on the bottom of the top plate to tilt the sides of the top plate up or down. The geometry of the wedges and the pusher can be modified to increase or decrease the plate's rotation about the mechanism's axis. The top plate has screw holes for attaching the spinning mechanism to the tilting system.

Initially, this design implemented two vertical linear actuators on each side of the top plate to push them up or down. However, the nominal height of the linear actuators without extension takes up more space than desired for the overall tilt plate design. The taller the

mechanism, the more unstable the attached spinning system would be. Thus, the team decided not to continue with the vertical linear actuators for tilting the top plate.

The early prototype tilt plate design was created before experimental testing to decide the required angle for adequate wetting was conducted. The design assumes that the required angle for wetting is substantially small compared to the current manual method of placing the bioreactor onto one of its corners (90° tilt along the long axis). The procedure used to test the minimum required wetting angle can be found in the [linked document](#) and the angle was found to be 15° which this design would be capable of providing.

Current Design



Fig 8: (Left) The current tilt plate design uses a horizontal lead screw to push the top plate up at an angle. The motion would be constrained by the height of the pusher and the hinge attaching the top and bottom plate. A straight section of the bottom plate would be removed and a section of the top plate will be removed at a downward angle such that at the beginning of the pusher's path, the top plate sits flat and as the pusher moves forward the space provided for the spinner decreases and the pusher lifts the top plate up. (Right) The mechanism would be used in tandem with the Spinner design alternative to provide a second, more constrained axis of rotation for wetting. The Spinner mechanism would be attached on top of the tilt plate such that the Spinner would lift at a tilt with the top plate of the mechanism.

The current design suggested for the tilt plate mechanism attempts to solve the issues of the early prototype in manufacturability and balance. The design is inspired by a door hinge, using a top plate and a bottom plate connected by a hinge such that the top plate is free to move up and down. The mechanism makes use of a pusher that consists of a spherical top and bearing on the bottom to reduce friction when moving. The pusher is moved forward by a horizontal lead

screw. As the pusher follows along the straight slot in the bottom plate, the downward slanted slot in the top plate means that the space for the pusher decreases as the pusher moves forward and the top plate is forced to lift up to accommodate the set height of the pusher. This also means that the closer the pusher is to the hinge, the higher the top plate will lift. In this way, the mechanism will be able to tilt the spinner at the required 15° .

For wetting, the Spinner will rotate the bioreactor to 75° along the short axis and then the tilt plate will engage to tilt the bioreactor 15° along the long axis. The Spinner will remain on the tilt plate and will sit flat when not wetting.

Failure Stop

Ultimately the design must fail predictably and therefore fail in a way that maintains the position at the point of failure. To ensure the robotic system remains safe and immobile during unexpected power loss or control failure, a passive mechanical stopping method was explored. Among several concepts, a wheel-brake or ratchet-claw mechanism emerged as a promising solution. This approach offers a fail-safe response—automatically engaging when power is removed—and provides a simple, robust means of preventing unwanted wheel rotation. The following section describes the operation, advantages, and anticipated integration challenges of this mechanism.

Wheel Brake

In order to accomplish this failure stops must be implemented to prevent the system from spinning or continuing

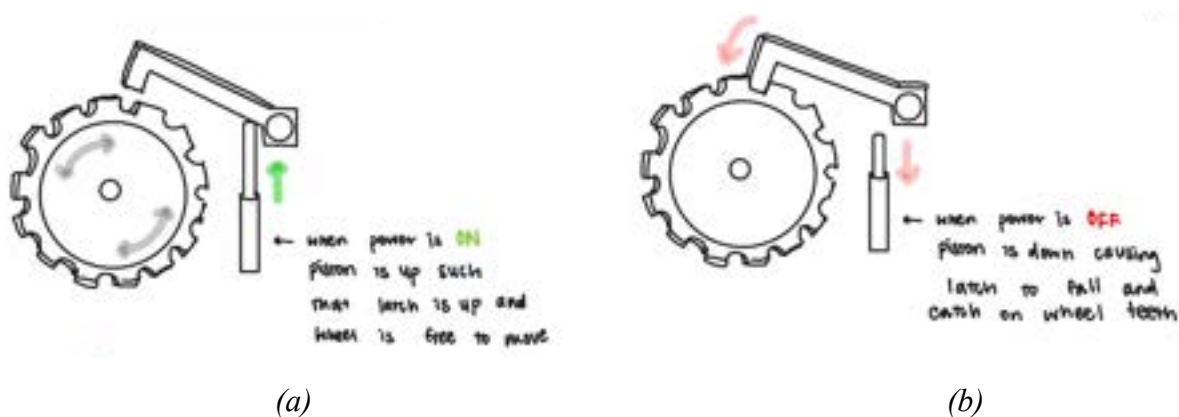


Fig 9: The wheel brake mechanism uses a piston that (a) holds the brake up when power is on and (b) releases the piston upon power failure or system malfunction, such that the brake falls into the groove of the wheel. The wheel's teeth and the shape of the brake's "beak" prevent the wheel from moving in either rotational direction.

The wheel brake or ratchet claw mechanism was considered as a possible design to address power failure or system malfunction. The design is inspired by an emergency car brake that does not allow the wheel to move forward or backward when the brake is installed. The mechanism works using a piston that holds the brake/claw up while power is on (see Fig. 9a). If the power fails or there is a system malfunction, the piston drops the claw into the wheel (see Fig. 9b). Due to the square divots in the wheel and the square tip of the brake, the wheel will not be able to move in either rotational direction when the brake is in place. After evaluation, the design was deemed simple and effective enough to be moved forward into the prototyping stage. Foreseen challenges include the positioning of the brake system on the robotic system and the availability of a wheel with the same square or possibly even triangular teeth that is either the motor wheel or able to attach to the motor that is moving the system.

Fixture

The fixture designs were developed with the Spinner design in mind. The purpose of the fixture is to attach the bioreactor to the driving shaft and free-spinning axle in the Spinner design.

The most probable point of failure for these fixture designs would be the fixture breaking under the stress induced by the system's vibrations. The intention for the fixture system is for it to be reused for each run. Each of these fixture designs rely on their geometry to rigidly constrain the bioreactor. Since a vibration system will be used on the bioreactor, any rigid fixture system will experience fatigue over many uses. When the vibration (and damping) system is designed, the fatigue stress can be determined, and then the dimensions of the geometries of each fixture can be determined based on their abilities to sustain the applied fatigue stress.

This section is a summary of the fixture ideation and final prototype. Refer to Appendix G for a detailed walkthrough of earlier fixture ideas.

Early Fixture Design - Slider with Pin

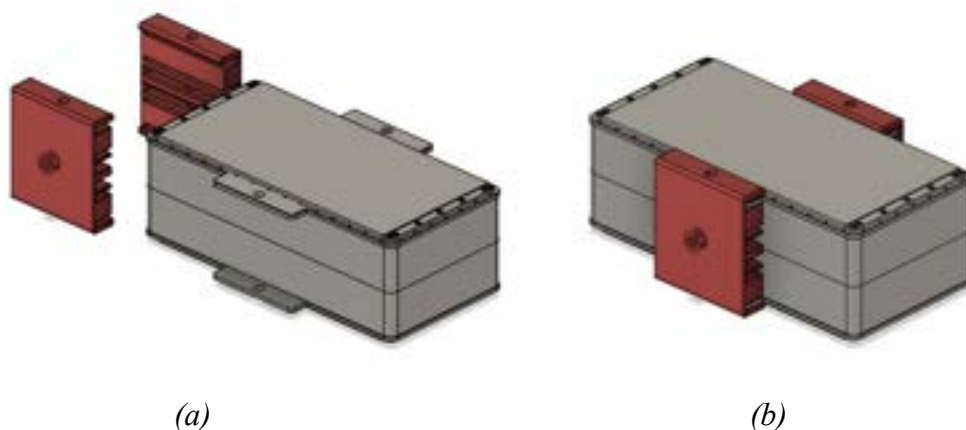


Fig 10: (a) The Slider and bioreactor before connection are depicted above. The rectangles extruding from the top and bottom planes are a part of the top and bottom platens. (b) The bioreactor extrusions fit into the Slider slots as shown above.

The Slider's purpose is to attach the bioreactor to the shafts. The shafts are secured to each Slider by a set screw. The platen will have four extrusions as seen in Figure 10a, which slide into the corresponding slots of the fixture as seen in Figure 10b. The platen extrusions and Sliders each have a hole running top to bottom through the center. Once the bioreactor slides into the fixture slots, a pin will fit into the slot to secure the Sliders to the bioreactor.

Bioreactors will have different heights depending on the number of membranes inside it. A major advantage of the Slider is its ability to accommodate all existing membrane stacks without needing to be switched for different-sized fixtures. The Slider was designed to accommodate stacks of 5, 10, 20, and 50 membranes. These are the stacks XDemics plans to make commercially available. It should be noted that the Slider requires XDemics to change the platens to include the four extrusions that slot into the fixture design.

Between latches and pins for securing the fixture to the bioreactor, pins are advantageous because they are easier to design and manufacture to fit the Slider, and pins are simpler to work with for gloved hands.

A disadvantage of the Slider lies in how it is set up. One of the Slider fixtures is attached to a motor, which will keep it in place when the machine is off. The other fixture is attached to a free spinning axle. This means that during the set up of the bioreactor, the technician will have to hold the fixture with one hand, and slide the bioreactor in with the other hand. Other fixture designs allow for the technician to hold the bioreactor with both hands during set up. A possible remedy to this is adding a beam that connects the bottom of both fixtures, so that the slots are aligned for set up.

Currently, the platen extrusions are designed with the arbitrary dimensions 5" x 1". The dimension for the extrusions can be adjusted based on the expected forces and for ease of use.

Final Fixture Design

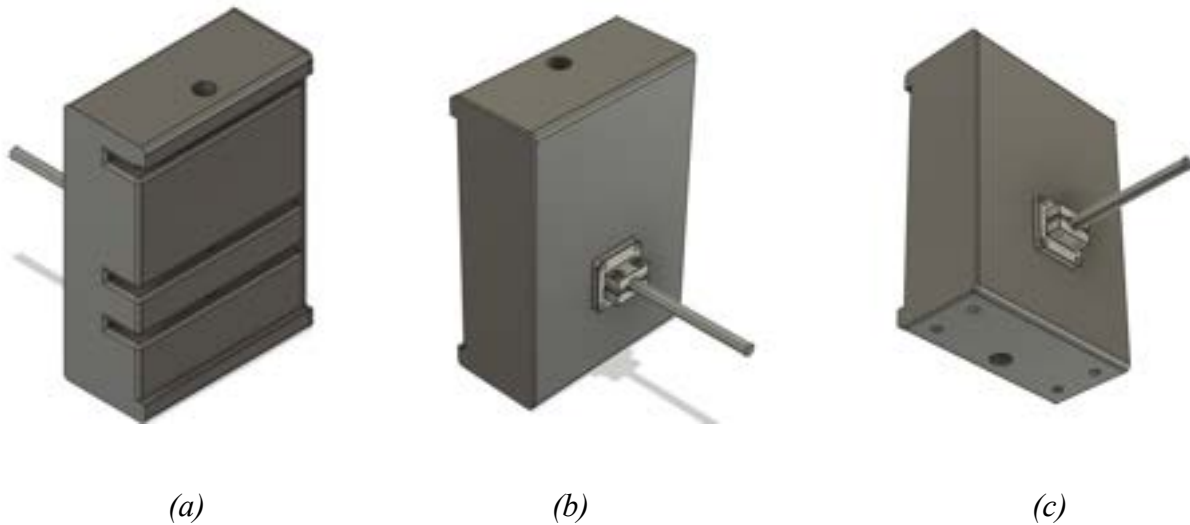


Fig 11: Rendering of the final fixture in Autodesk Fusion.

The final design of the fixture is intended to attach the bioreactor securely to the shaft. The final design includes extra slots that couple the vibrator close to the bioreactor, allowing for

minor displacement and repeated impacts. Key design elements of the fixture include a pin slot, vibrator platen slot, backing, bioreactor slots, and an attachment point for the connecting piece.

A major strength of this design is its potential for scalability. The basic fixture design, such as the part shown in Figure 11a, is suited for a bioreactor with five layers, but the design can be scaled up to accommodate larger bioreactors. The design shown in Figure 11b, featuring additional slots, demonstrates that the fixture can accommodate bioreactors with varying numbers of layers. This durability has been confirmed by FEA Analysis, as the fixture is designed to support a 50-stack weighing up to 60 lbs.

For ease of use, the fixture is designed to be easy to handle with gloves and requires no tools for setup. While the prototype uses two bolts and wing nuts, the plan is to replace these with clevis pins, which can be attached and removed by hand. Consideration must also be given to how vibration transfer changes with different bioreactor thicknesses.

Regarding materials, the discussion focuses on using stainless steel for parts to minimize particle generation, simplify manufacturing, and comply with cleanroom standards.

Manufacturing Recommendations for Fixture

[illegible]

Fig 13: CAD Drawing of Platen extrusions. This is a modification of a current XDemics design.

This section outlines the engineering decisions and specifications regarding the clearances and tolerances for the bioreactor's slot dimensions and positioning. These choices ensure optimal mechanical alignment and energy transfer during operation. The distance between the top two slots is dictated by the specific platen-vibrator distance required for optimal vibration transfer. The tolerance for this dimension is carefully calibrated based on the range of vibration transfer needed to agitate the cells effectively within the designated operational specifications.

Top Slot Height

A wide tolerance is preferred for the top slot height, as it can positively influence the transfer of vibration throughout the system. To accommodate this requirement and allow for sufficient clearance, a loose running fit (H11/c11) is implemented. This fit prevents mechanical binding while facilitating efficient energy transmission.

Bottom Two Slot Heights

To ensure the structural integrity and stability of the lower section of the assembly, the bottom two slot heights are manufactured with a close running clearance fit (H8/f7). This precise fit minimizes unnecessary movement and maintains the necessary alignment within the system.

Distance Between Bottom Two Slots

The locations of the bottom two slots require a tight tolerance because they are determined by the overall bioreactor geometry, and the slots themselves are long. Maintaining this tight tolerance is critical for proper assembly and function. Therefore, a close running clearance fit (H8/f7) is also applied here to ensure high accuracy and a secure, predictable mechanical fit.

Vibration Mechanism

In order to couple a vibration source with the mechanism, several design approaches for integrating the Eccentric Rotating Mass (ERM) vibrator were considered. The team first evaluated mounting the vibrator directly to the base structure and allowing vibration to propagate

through the supporting stand and fixturing. Another approach considered attaching the ERM directly to the bioreactor plate itself, such that the vibrator would either become a consumable component attached to each bioreactor or could be inserted through an interlocking rail system. While these solutions demonstrated potential, they introduced unnecessary complexity and raised concerns regarding manufacturability, maintenance, vibration losses, and scalability.

The team then realized that the same sliding fixturing mechanism used for constraining the bioreactor platens could also be adapted for the vibrator itself. By mounting the ERM to a dedicated vibration platen and incorporating an additional slot into the fixture design, the vibrator could be securely integrated into the system while remaining modular and removable. This approach allowed the vibrator to directly interface with the bioreactor while still being mechanically isolated from the majority of the supporting structure.

The resulting coupling mechanism uses a guided sliding interface that constrains the vibration platen within the fixture while allowing limited oscillatory motion in the desired direction of vibration. During testing, it was found that tightly constraining the vibrator against the bioreactor reduced vibration transfer efficiency due to damping effects. Introducing a slightly looser fit between the vibration platen and fixture allowed the ERM to oscillate more freely and repeatedly impact the bioreactor surface, substantially improving vibration coupling. Experimental measurements showed that this modification increased transmitted vibration amplitudes from approximately 6–7 mm/s² to nearly 30 mm/s² at the 24-well plate surface.

This design also minimizes the transmission of vibration into the surrounding rotational system, reducing fatigue loading on the fixture and support components over repeated operation cycles. Additionally, the modular sliding mechanism allows the vibration system to be easily removed, replaced, or scaled for future bioreactor configurations without requiring major modifications to the primary fixturing geometry.

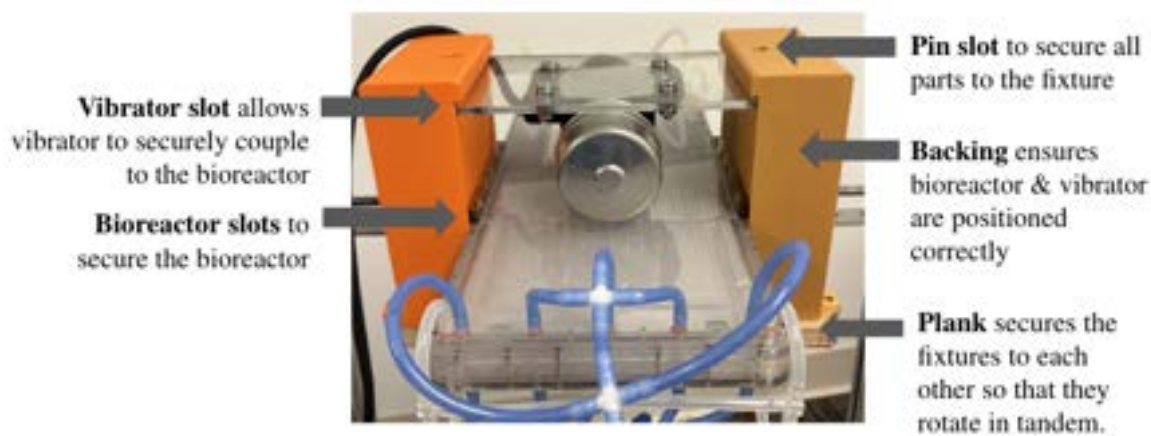


Figure 14. Complete fixture assembly prototype with Netter vibrator slotted in and coupled to bioreactor.

Tube Management

Connected to the bioreactor are tubes serving as the inlets and outlets of the system. Without consideration of tube management, the tubes are then free to move and could get stuck while the bioreactor is rotating. Fluids are also flowing in these tubes, so the risk of the tubes getting stuck increases the risk of interfering with fluid flow into and out of the bioreactor. Each side of the bioreactor also has a group of tubes coming from the top and bottom of the bioreactor. Having many tubes increases the chances of entanglement and disrupting the rotation of the bioreactor. Given that the bioreactor must rotate on multiple axes, it is then important to consider how to manage and route the tubes to avoid potential entanglement and excessive pinching.

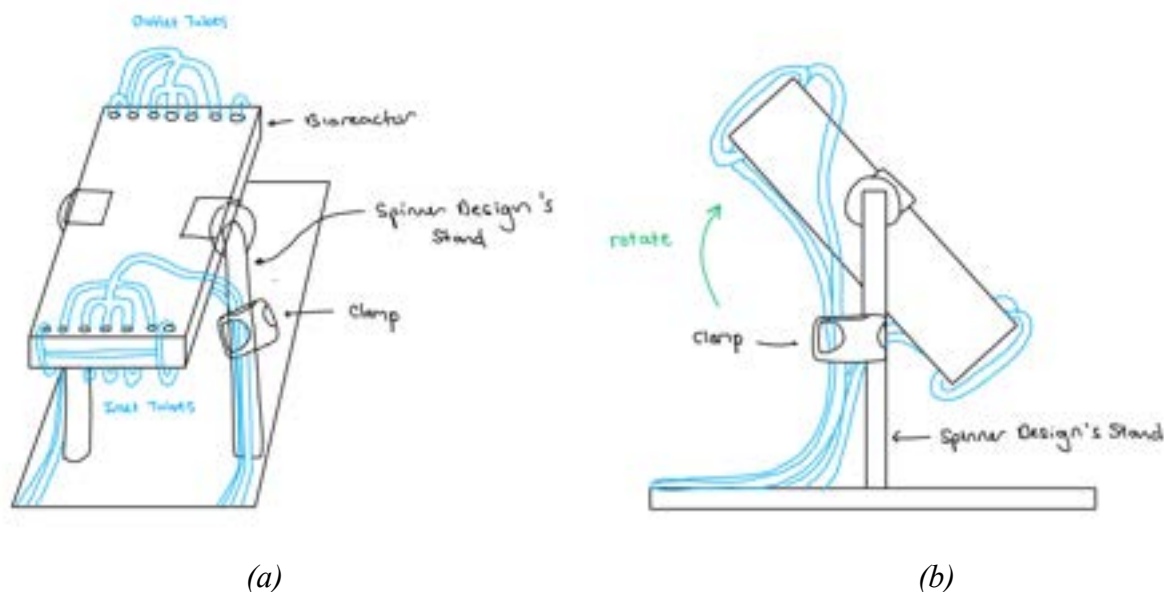


Figure 15: (a) Isometric view of a clamp with tubes. In this figure, the rigid body the clamp is attached to is the Spinner design's stand. On each stand, a clamp is attached to hold the bioreactor's inlet or outlet tubes. The rest of each tube's length is then routed along the sides of the bioreactor and system. (b) Side view of the routing of the tubes when held by the clamp. With long enough tubes, there is enough slack to move with the bioreactor's rotation.

A proposed design to handle tube management is a tubing clamp. This clamp is attached to a rigid body of the overall system, such as the stand from the Spinner design. The clamp then holds a section of the tubes together while the rest of the tube's length is routed along the sides of the bioreactor during rotation (see Figure 15a). With this routing, the tubes move with the bioreactor's rotation rather than the bioreactor pushing against the tubing during rotation, since the tubes are long enough to provide enough slack for routing during rotation (see Figure 15b). This design includes two clamps, one for the inlet tubes and the other for the outlet tubes. Advantages of this design include simplicity, affordability, and effectiveness in reducing tube tangling and pinching. However, disadvantages include potential particle generation and overtightening of the tubes such that they restrict fluid flow. If the tube is restricted and the fluidics of the bioreactor are not properly performed, the viability of the cells may be compromised, depending on the severity and duration. To address this issue, clear user instructions for the tubing clamp will be provided with this design. The issue can also be

addressed by considering additional design alternatives that do not require an adjustable clamp and instead a fixed size for the tubes to be held by.

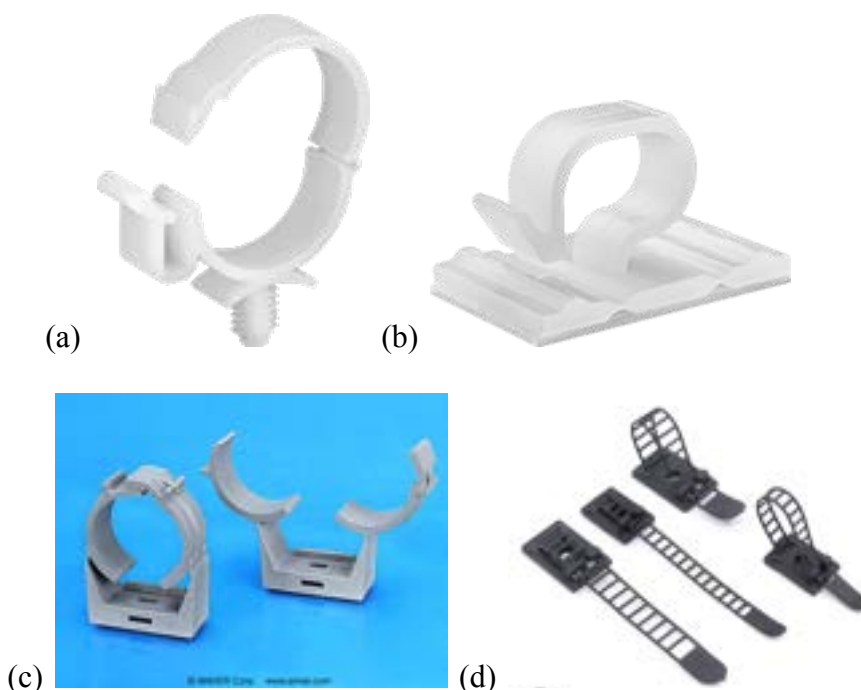


Figure 16: (a) Example cable holder style from McMaster-Carr with a push-in mount. [9] (b) Example cable holder style from McMaster-Carr with a slip-in mechanism. [9] (c) Example of a clamp with a hook to open and close the clamp. [10] (d) Example of adjustable cable straps that have an adhesive back mount. [11]

Additional tube management design alternatives include variations on how the design is attached to the rigid body of the overall system and how the tubes would be held. Examples of different attachments include an adhesive back mount and a push-in mount (see Figure 16a). However, an adhesive back mount should be avoided since the cell seeding and cell harvesting using the bioreactor would be in a clean room. The push-in mount can be used by adding a hole to a rigid body, such as a hole in the Spinner design's stand, and inserting it in. An advantage of this mount includes being a more mechanical way for attachment that better suits a clean room. Moreover, instead of tubing clamps, tubes can also be held by a slip-in mechanism, a clamp closed with a hook, and adjustable cable straps (see Figure 16b-d). The slip-in mechanism is inspired by medical IV tube line organizers, where fluids flow through the tubes and tubes are held by slipping them into an opening that leads into a closed space for the tubes to be held in. However, with the bioreactor rotating, it is important to consider the chance of tubes being

pinched at the opening. A clamp closed with a hook is another design where the tubes can be easily held, and the hook provides a means of holding the tubes in place. A potential challenge includes finding a diameter size of the fixed clamp that can hold a bundle of tubes simultaneously. Compared to the tubing clamp, this clamp reduces the chances of pinching tubes because they can be placed in the clamp by opening the clamp wide and closing it with a hook. As for the adjustable cable straps, various-sized bundles of tubes can be held together and secured in place. They are also an alternative to velcro-straps, but are more suitable for clean rooms. However, a challenge would be finding a strap that does not have an adhesive back mount. An advantage of all these alternatives over the adjustable tubing clamp is that they reduce the chance of pinching the user's gloves in a clean room while still holding the tubes securely.

Vibration Design Alternatives

Some mode of vibration is needed to dislodge the cells from the wells during cell recovery, however, the most effective frequency, magnitude, and direction is unknown. Before the team's site visit, several modes of vibration were ideated. However, after the site visit the team and liaisons decided it was best to conduct vibrational testing before choosing a design alternative. This section will detail the alternatives ideated prior to the site visit.

The functions that agitate the bioreactor will be integrated into each of the designs with modifications. A common vibration mechanism used in designs are ERM (eccentric rotating mass) motors that use a weight attached off-center on the shaft of the motors. A vibration is then produced when the motor spins. The faster the motor spins, the stronger and higher frequency the vibration is. There are also smaller versions of ERM motors called coin vibration motors. They use an off-center rotating mass, but they are cylindrical and flat, allowing them to be mounted to flat surfaces. These small coin vibration motors could be placed on the surface of the bioreactor and then produce vibrations needed to dislodge the cells. However, this idea was deemed not viable due to requiring a stronger vibration to dislodge the cells for cell harvesting. A scotch yoke is another mechanism as it turns circular motion into linear motion. This could be used to produce a quick back-and-forth motion applied to the bioreactor in place of vibration. A microfluidic solution is also possible. By altering the fluid and gas pressures in the bioreactor, it

is possible to dislodge the cells from the wells using a microfluidic approach. An additional approach is microfluidic acoustics. The process works by turning on low-frequency ultrasonic piezoelectric transducers, which gently shake and loosen the cells. This creates standing sound waves with high- and low-pressure points, called antinodes and nodes. The cells naturally move toward the low-pressure nodes, where the agitation helps detach them. Pumps are then used to control how the cells flow through the system. This method allows for cell viability as the low power density of acoustic waves are safe for cells, as well as scalability for high throughput cell processing. However, this brings a challenging microfluidics approach for vibration since the layer of membranes in the bioreactor would likely dampen the acoustic waves.

A major concern with vibration is how it will couple with the rest of the system, possibly inducing stress to the other fixtures causing failure over time. One method to avoid this is to decouple the vibration from the rest of the fixture through the clamp. For example, a two-layer frame shown in Figure 15 consists of an inner layer that attaches directly to the bioreactor and then an outer layer that would interface with the mechanism used to rotate the device. Between the two layers, the vibration mechanism is housed so that the vibration is isolated from the fixturing necessary for rotation. Alternatively, the bioreactor could be completely removed from the rotational system via some automated removal mechanism. Each method would effectively decouple the vibration from the rest of the system avoiding stress of the rest of the system.

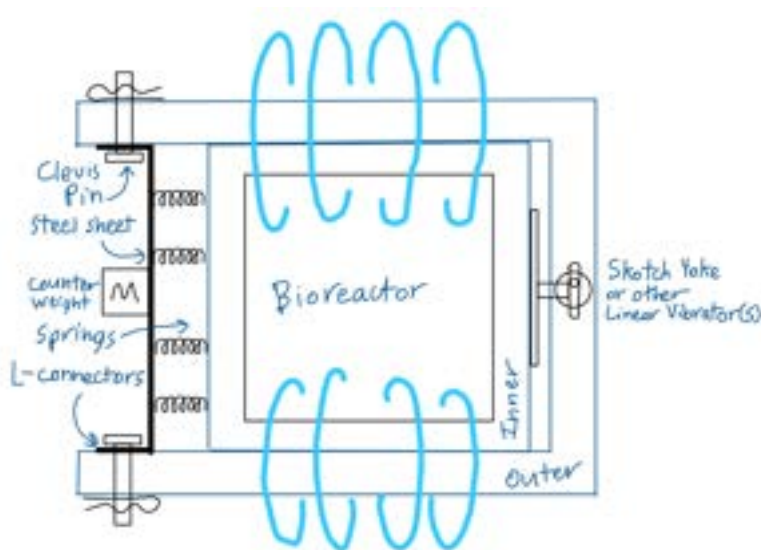


Figure 117: Two-layer frame design with inner and outer layer separated by vibrational mechanism. This mechanism would allow the bioreactor to be agitated without applying excess stress to the surrounding mechanisms and fixturing.

Experimental Work

The experimental work section contains procedures and tests for experiments that aided in understanding the limitations of the proposed system and in modeling certain specifications of the system. Included in this section are the COMSOL simulations of flow across a 2D cross section of the bioreactor to analyze the shear stresses on the system when the bioreactor is in motion and the manual vibration testing with microspheres and a 24-well plate to assess the necessary force required to dislodge the microspheres from the silicone grooves.

2D Cross Section COMSOL Simulation

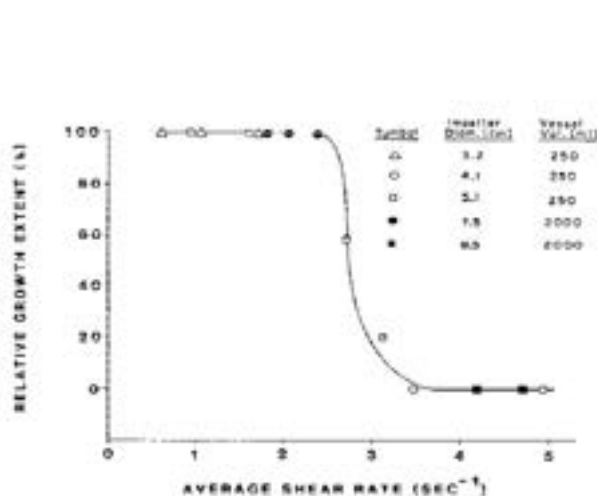


Figure 5. Relative growth extent of FS-4 cells on microcarriers as a function of the average "time-averaged" shear rate in the outer radial region. Results are calculated from the integrated shear factor data of Hu, Doctoral dissertation, M.I.T., Cambridge, MA (1983).

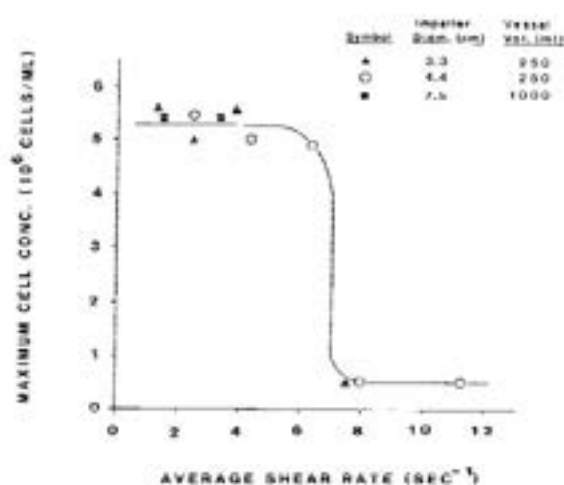


Figure 6. Maximum cell concentration for chicken embryo fibroblasts grown on microcarriers as a function of the average "time-averaged" shear rate in the outer radial region. Results are calculated from the integrated shear factor data of Simsky, Melchaker, Tyo, Giard, and Wang, *Ann. N.Y. Acad. Sci.*, 369, 47-59 (1981).

Figure 18: Effects of shear stress on CHO cell growth, in a bioreactor. [13]

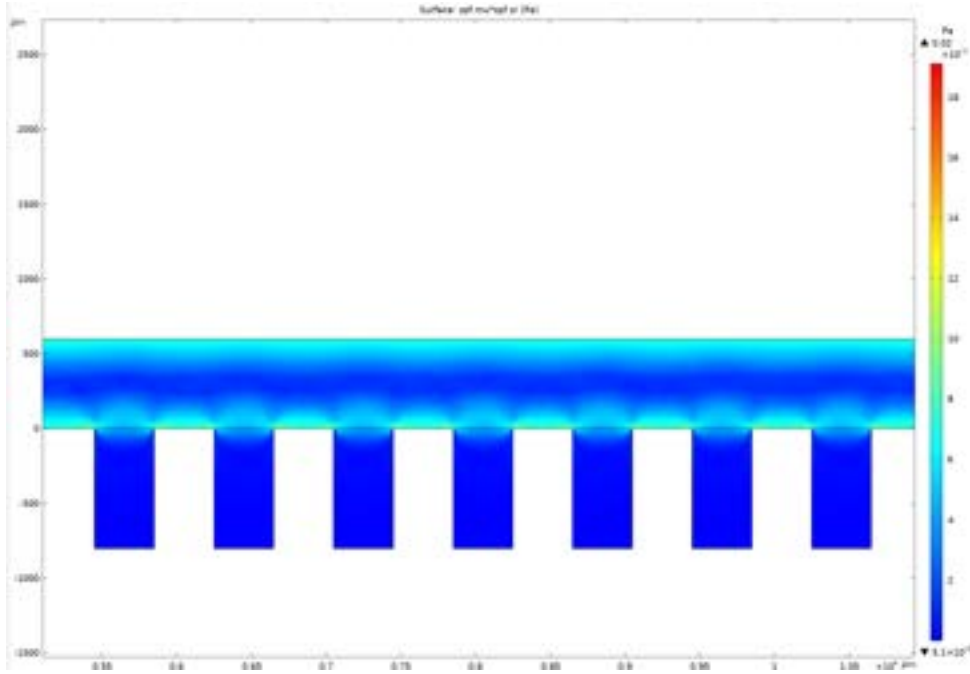


Figure 19: CFD: Plot of shear stress in the cross section of wells in a cell membrane, at 5mL/min/membrane. Cells experience this shear stress below the channel in the wells.

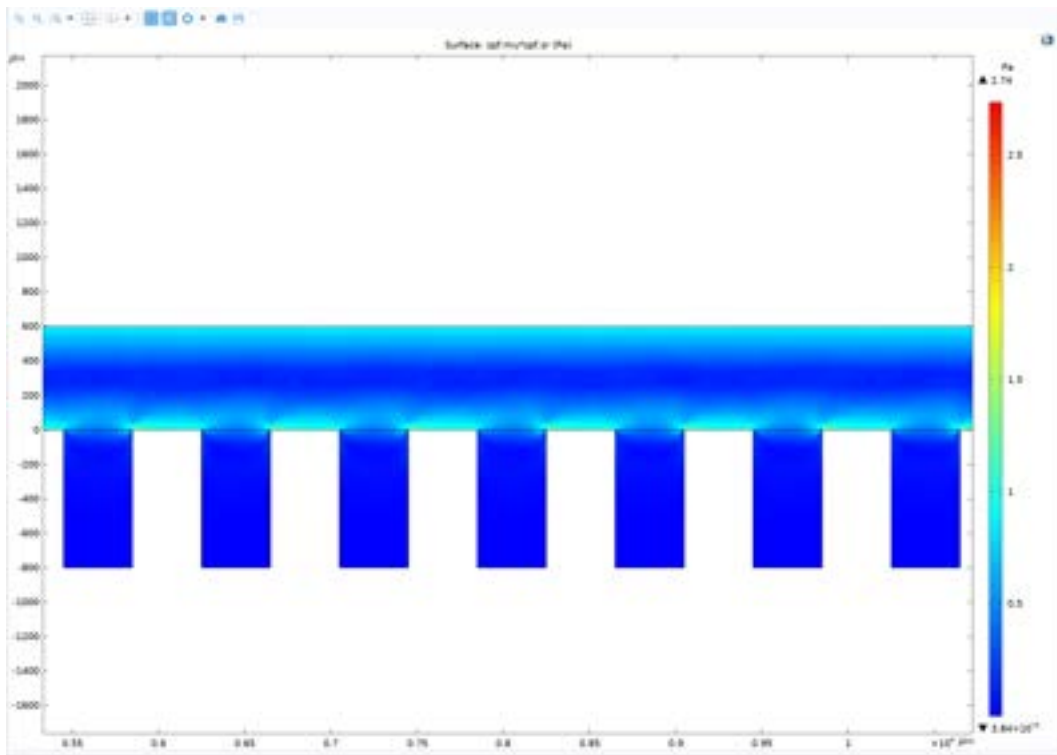


Figure 20: CFD: Plot of shear stress in the cross section of wells in a cell membrane, at 600 mL/min/membrane. Cells experience this shear stress below the channel in the wells.

Cells in the bioreactor grow best in specific conditions. When the fluidic pump system is developed in the Spring, an important consideration will be the shear stress inside the cell membranes. For the cells to adhere to the wells and multiply, they cannot undergo too high of a shear stress. The shear stress equation is $\tau = \mu$ (dynamic viscosity) * γ (shear rate). A shear rate (γ) > 8 s⁻¹ will significantly hinder CHO cells from growing (Fig. 18). Assuming the μ (dynamic viscosity) of the cell media is 0.95 mPa*s, and the ρ (density) of medium is 993.3 kg/m³, the shear stress inside the wells cannot exceed 8 mPa.

The cell membrane must be greatly deformed in order for the shear stress to be consistently above 8 mPa inside the wells. According to Fig. 18, a shear stress over 8 mPa could dislodge the cells.

Fig. 19 displays the results of a COMSOL simulation on a cross section of the cell membrane to find the shear stresses. In this study, the inlet flow was assumed to be 5 mL/min/membrane. The conversion from “mL/min/membrane” to m/s is the following equation:

$$\frac{mL}{min} * \frac{10^{-6} m^3/mL}{60 s/min} * \frac{1 \text{ membrane width}}{0.16764 m} * \frac{1 \text{ membrane height}}{0.0006 m} = \frac{mL}{min} * \frac{1 min}{6035.04 mL} = \frac{m}{s}$$

The shear stress inside the wells is about 2*10⁻³ mPa. This verifies that the CHO cells inside the wells will be able to grow with a medium velocity of 5 mL/min/membrane.

Using the undeformed geometry of the cell membrane, it is impossible to use fluid velocity to dislodge the cells. Fig. 18 is from a study with inlet velocity 600 mL/min (0.1 m/s), and almost half of a given well has a shear stress below 8 mPa.

Simulation	Purpose	Relevance
1. COMSOL 2D - Fluidic resistance of 5-stack membranes	Find fluidic resistance of a cell membrane. If it is low, there will be a uniformity of flow between membranes. Ensure flow between membranes is within 5-10% of flow rate of other membranes.	As we scale to 50 membranes will the same amount of fluid be reaching each layer? Will mixing occur evenly? Determines the effect of hydrostatic pressure on the system.

2. SOLIDWORKS FEA - membranes stacked together	Modeling deformation based on pressure inside the gas membrane, in an overpressurized scenario.	Needed for designing gas membrane for cell agitation. How will the membranes deform after the gas membrane is over pressurized?
3. COMSOL 2D - gas membrane deformation	Modeling gas velocity needed to create the desired gas membrane deformation.	Compliments with SOLIDWORKS study for designing gas membrane cell agitation.

Table 5: Chart of planned fluid and solid mechanics simulations. These simulations will provide insights into the forces inside the bioreactor and how they might impact cell viability.

Additionally, using vibrations in the gas membrane could agitate the cells in the bioreactor, as an alternative to mechanical vibrations. The future steps in investigating this possibility are to determine the required values of three variables which relate to shear stress (inside the wells) at 8 mPa. These values are inlet velocity of the cell membrane (scalar), pressure in the gas membrane (gradient field), and deformation of the cell membrane (deformation vector field). Simulations 2 and 3 in Table 5 describe two complementary studies to determine these values.

Vibration Testing

After the team's Fall Semester site visit, the team and liaisons decided that the method for vibration needs to be proved through testing. The direction in which the bioreactor is vibrated needs to be determined through empirical testing. To do so, manual vibration by shaking with hands and higher frequency vibration using a massage gun will be used to create linear motion in the x,y, and z. This method will be tested on microspheres using the 24-well plate. A sweep of direction and varying frequencies in each direction will determine which direction, frequency, and magnitude is the most effective for dislodging the microspheres. Based on the results of this testing, the appropriate vibration mechanism will be integrated into the prototype.

To test the viability and required vibration for the system, small-scale testing was conducted. The goal of this testing was to experiment with different vibration techniques to determine the best way to implement vibration in the prototype, including the direction and frequency of vibration. The small-scale testing was performed on the 24-well model to simulate

the large bioreactor. Plastic microspheres sized 10–90 μm were used to most closely simulate the size of Chinese hamster ovary (CHO) cells. The experimental procedure found to be most effective was as follows.

Materials and Equipment Needed:

- Dionized (DI) Water
- Dish soap
- Pipette and pipette tips
- 10–90 μm plastic microspheres
- XDemics 24-well plate
- Glass slides (or any small sheet of glass)
- Isopropyl alcohol 90%
- SYLGARD 184 kit
- C-Clamp
- Heat gun and/or oven
- Huber needle and syringe
- Massage gun (2000-2800 rpm range)
- Microscope

Setup & Sylgard Procedure:

- Wet each well with DI water and detergent mixture using a pipette to remove any bubbles
- Suspend the microspheres in the same mixture in a separate container
 - Suspension can be done by repeatedly pipetting the mixture and microspheres until the microspheres appeared to be suspended properly
- Pipette 0.5 mL of suspended microspheres into each well
- Each well should be filled a little less than halfway
- Prepare surfaces for sylgard curing
 - Wash the glass slide with detergent and water and then wipe down with 90% isopropyl
 - Clean the top of the silicone wells where the glass will be adhered with 90% isopropyl alcohol
- Now, prepare the sylgard in a glass container

- 10:1 mixing ratio
- Mix with a glass rod
- Try to avoid air bubbles while mixing
- Apply the sylgard to the top of the silicone wells, ensuring an appropriate amount to adhere and not leak into the well
- Once the sylgard is applied to the silicone, place the clean glass slide on the area and gently clamp with a C-clamp to the table (see Figure 16).
- Allow the Sylgard to cure
 - Leave at room temperature for a few days, minimum 34 hours
 - Heat up the sylgard in an oven at ~150-200F (<212F to avoid water boiling) for 10-15 minutes
- Allow the silicone and glass to cool before proceeding with testing



Figure 21: Silicone plate and glass clamped for the Sylgard curing process.

This photo depicts the 24-hour curing process of the glass slide in order to seal the microspheres and deionized water mixture for testing.

Vibration Testing Procedure

- Fill up the wells completely with DI water using a needle and syringe.
 - Enter from the side of the well to add DI water to the well. You must also use the syringe to remove the excess air in the well.
 - Add water, remove air, and repeat this until the well is completely filled with liquid.

- Document the state of the wells by using a microscope to capture the microspheres at the bottom. This is done best by capturing from the bottom of the well.
- Start manual vibration techniques.
 - Invert the bioreactor and shake either side to side or up and down (depending on the desired experiment).
- After the manual vibration, document the state of the wells. Note any observations in how many microspheres were dislodged from the well during the manual movement.
- Next, you may apply high-frequency vibration using a massage gun.
 - Experiment with force applied to the glass, bottom of the wells, sides of the wells, and varying frequencies of the massage gun (see Figure 17).
- Document the state of the wells using the microscopes. Note changes in the amount of microspheres still stuck in the wells (see Figure 18).
- The plates can then be left for the microspheres to settle in the wells and be reused for experimentation.

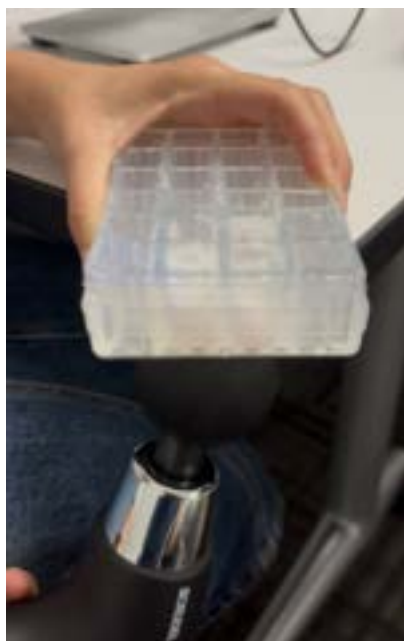


Figure 22: Example of vibration technique used with massage gun. A 24-well plate was hand held in place and the massage gun held on the underside of the plate allowing displacement on the opposite end of where the plate was held. The massage gun was held at a 90 degree angle and the cells were agitated for 1-2 minutes using the vibration of the massage gun before being inverted.

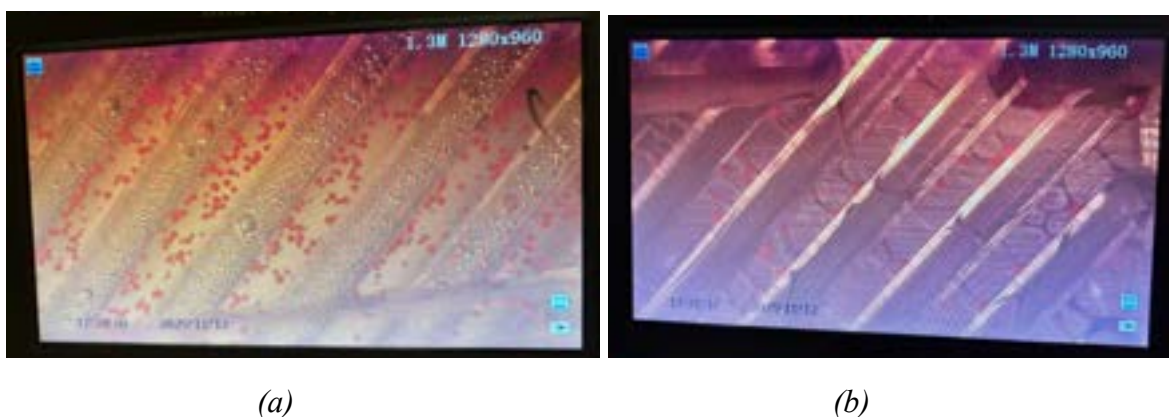


Figure 23: Before (a) and after (b) of massage gun vibration technique applied to silicone wells. These images show (a) the concentration of microspheres in the wells before agitation, and a reduced concentration after agitation. These images were taken on a microscope and show at least 80% of microspheres dislodged.

This procedure was iterated throughout the semester to determine the best way to conduct the testing. Most delays resulted from ordering needed materials. Using this procedure, the team found that a detergent was required to achieve any dislodgment of the microspheres. It was also found that manual manipulation was not enough to remove all microspheres from the wells; it removed only about 50%. This meant that vibration using the massage gun against the glass with the plate inverted was the most effective method for removing the microspheres. However, even the most successful trials did not achieve 100% removal. Possible reasons for this include the amount of detergent used, since adding detergent improved dislodgment; more may be needed for better results. The detergent used was basic dish soap, so a detergent designed for cell experimentation or a specific medium might be better suited for this work. The microspheres may also be adhering to debris in the wells because the wells were not fully cleaned or sterilized when used. This issue would be eliminated in an actual bioreactor designed for experimentation.

For wetting techniques, the team found that pipetting with a detergent–DI water mixture was the most effective way to ensure the wells were properly wetted. However, this mixture made it easier for bubbles to form when there was less liquid in the wells. The desired amount of mixture is about one-third of the well, which helps ensure that the pipette tip stays fully submerged during pipetting.

Wetting Angle Testing

Wetting is the process of filling the bioreactor with a media to ensure each well is wet with no air bubbles. This ensures complete membrane surface coverage for maximum growth. Following discussions with the project liaisons, the team determined that the required tilt angle for a full scale membrane must be validated empirically. Small-scale wetting testing had shown to be successful with the use of 40% ethanol, however, full-scale membrane wetting with the HDCR geometry might respond differently due to factors that include channel height, membrane surface area, and flow path complexity. Therefore, systematic tilt testing was designed to determine the minimum angle required to achieve complete membrane wetting without trapped air.

The goal of this testing was to identify the lowest tilt angle that reliably produces a full membrane wetting with no persistent air bubbles that affect cell seeding. A perfusion system was used to introduce the 40% ethanol into a 5-membrane HDCR stack at controlled flow rates using a peristaltic pump. Angles of tilt, perpendicular to the axis of rotation for the HDCR, at 0°, 5°, 10°, and 15° were evaluated. To tilt the HDCR to these predetermined angles, we placed different sizes of wooden blocks under the left side of the prototype's frame, and ensured these angles were reached using an iPhone level in the Measure app. A test at 10° with vibration was also included since the vibration system was already coupled. The HDCR was also rotated to 75° from its flat resting position, with inlet tubes being at the lower end of the housing. To maintain this rotation angle, the HDCR was taped in place via strands of tape attached to the frame. Immediately following the completion of ethanol wetting, water was perfused through the system to clear the ethanol and minimize the swelling of the silicone membranes and tubing.

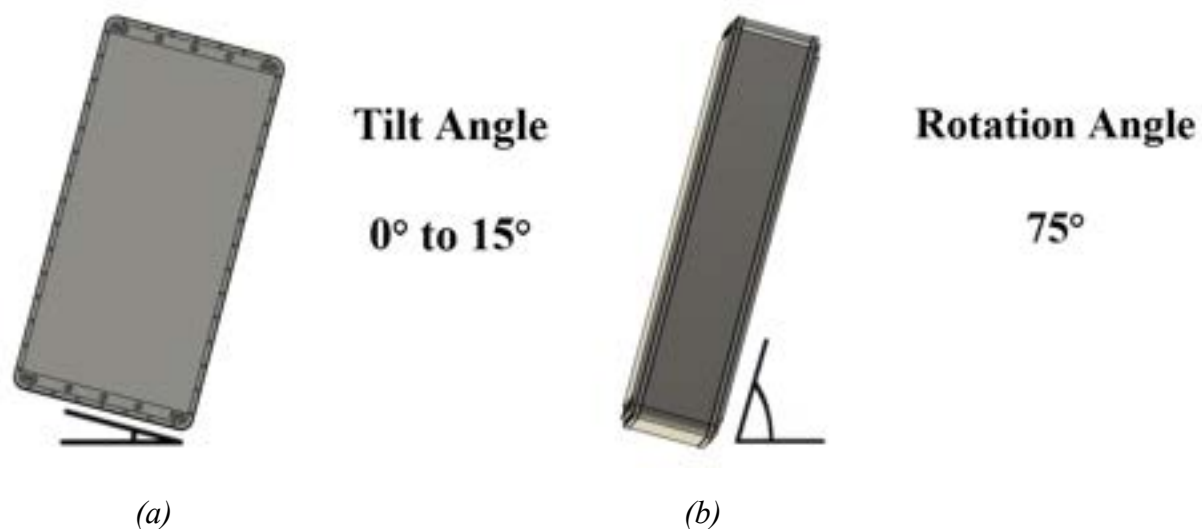


Figure 24: Diagrams representing the tilt (a) and rotation (b) of the HDCR that were performed during the wetting testing.

Materials and Equipment Needed:

- 40% ethanol solution
- Water
- 5-membrane HDCR stack
- Perfusion system (pumps, tubing, connectors, reservoirs)
- Blocks of different sizes to create required tilt angles
- iPhone level from Measure app
- Clamps or tape
- Absorbent pads or wipes
- Camera for documentation
- Waste collection container

HDCR Volume Calculations:

The HDCR fill volume was calculated as:

$$\text{HDCR volume (mL)} = 77\text{mL} + (60\text{mL} \times \text{number of membranes})$$

For a 5-membrane stack:

$$\text{HDCR volume (mL)} = 77\text{mL} + (60\text{mL} \times 5) = 377\text{mL}$$

This volume was used to standardize the perfusion quantities across all angle tests.

Wetting Angle Testing Procedure:

Setup:

- Assemble the HDCR and perfusion system according to standard operating procedures
- Verify all tubing connections are fully seated and clamped appropriately
- Place the HDCR onto the adjustable tilt fixture
- Set the angle to the desired test condition using a protractor
- Ensure absorbent pads are placed beneath the system in case of leakage

Wetting Angle Procedure:

- Prime the ethanol feed line until ethanol reaches the HDCR inlet
- Set the HDCR to the desired tilt angle (0° , 5° , 10° , or 15°) using wooden blocks placed under the left side of the prototype frame
- Begin perfusion of 40% ethanol at a low flow rate to minimize bubble formation
- Once the internal flow path is filled past all ports, increase flow to the standard testing rate
- Perfuse approximately 1–1.5 HDCR volumes of ethanol
- Assess wetting according to the WET-OK criteria:
 - WET-OK is defined as no visible macro-bubbles, no persistent microbubble clusters, a uniform continuous fluid film across all membrane surfaces, stability after a 30-second flow pause, and no additional intervention required to clear trapped air.
- If WET-OK criteria are met, immediately proceed to water perfusion.

Water Flush:

- Switch the feed to water immediately after WET-OK is achieved
- Perfuse a minimum of 2 HDCR volumes of water to clear ethanol from the system
- Continue perfusion for additional volume if ethanol odor or evidence of residual ethanol persists

Documentation:

- For each angle tested, document:
 - Actual measured tilt angle
 - Pass/Fail result
 - Observations regarding bubble location and behavior

The wetting angle testing demonstrated that as tilt angle increased, the ability to achieve complete membrane wetting did as well. When low tilting angles such as 0° and 5° were used for testing, air bubble production in the flow channels and membrane were significant, implying that full wetting had not been achieved. While performance improved further at 10° , inconsistencies were still observed during trials and adding vibration at this point saw an increase in the amount of bubbles formed in the HDCR. Once we increased the tilt angle again, wetting angle tests carried out at 15° saw full wetting taking place without any bubble formation, thus meeting the WET-OK requirements.

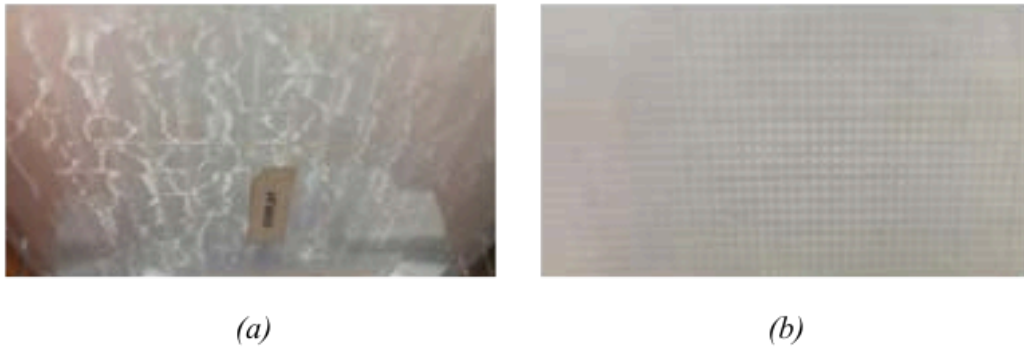


Figure 25: Comparison of HDCR membrane wetting at 0° and 15° tilt angles showing trapped air at 0° (a) and complete wetting at 15° (b)

As can be seen in Figure 25, comparing the visual outcomes at a tilt angle of 0° and 15° makes it clear how much better the wetting behavior is at the latter. The presence of bubbles is evident at a tilt angle of 0° , while there is no trace of bubbles at an angle of 15° , indicating that complete wetting is attained at this angle.

Wetting Angle Pass/Fail Table	
0°	Fail
5°	Fail
10°	Fail
10° with vibration	Fail

15°	Pass
-----	------

Table 6: Summary of wetting testing results across evaluated tilt angles.

The pass/fail test results as shown in Table 6 indicate consistent failures below 15° and success at 15°. Considering these results, a tilt angle of at least 15° will be included in the design of the tilt plate mechanism.

Vibration Testing (at XDemics Buck Institute Site)

Overall Procedure:

1. Attach the bioreactor and vibrator to the fixtures such that the vibrator is pressed up underneath the bioreactor (bottom face of the bioreactor) but has enough space to slightly oscillate up and down. Plug the vibrator into a power strip that is currently turned off.
2. Once the 24-Well Plate has been filled and capped, seal the edges and point of connection between lid and well with Parafilm until completely secure (no media spills out from the wells when rotated).
3. Stick a command strip onto the bottom of the 24-Well Plate along the long axis, then adhere the 24-Well Plate onto the top face of the bioreactor, approximately 1-2 inches from the front edge (edge facing you).
4. Tightly strap the 24-Well Plate to the bioreactor using a separate velcro band that runs around the long axis of the top of 24-Well Plate and the bottom face of the bioreactor.
5. Rotate the bioreactor 180° such that the 24-Well Plate is now upside-down
6. Turn on the vibrator for the allotted time by turning on the power strip. While vibrating, measure the vibration on the 24-Well Plate to ensure that the vibration is properly coupled.
7. When the allotted time is finished, turn off the vibrator by turning off the power strip and KEEP the bioreactor and 24-Well Plate UPSIDE DOWN. Hold the 24-Well Plate in place with your hands while carefully removing the velcro strap. Then, carefully remove the 24-Well Plate from the bioreactor KEEPING IT UPSIDE DOWN.
8. Visualize the disrupted and dislodged cells while upside down under a microscope.
9. Record approximate cell detachment percentage from observations.

Test 1: Control Test with Only Water

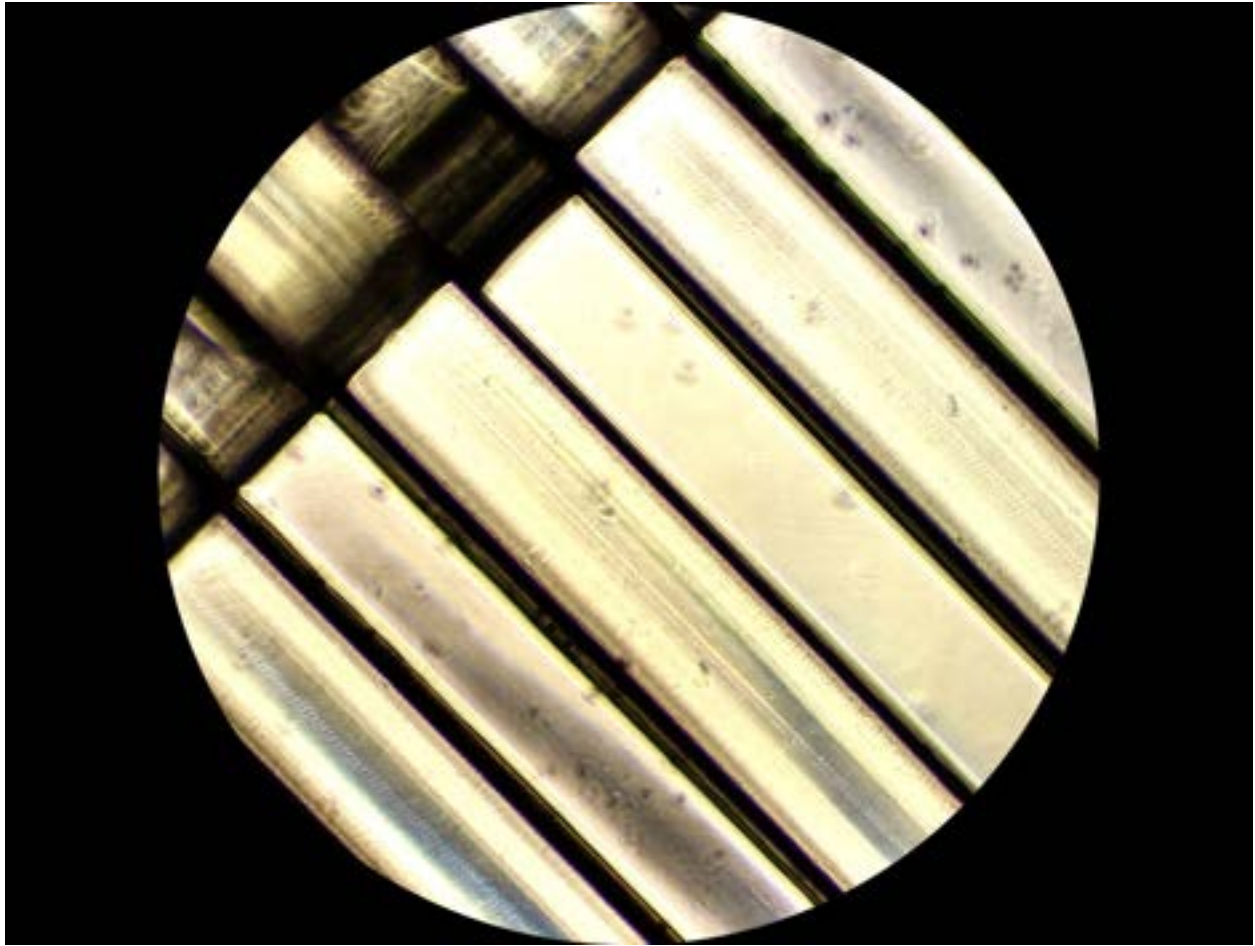


Fig 26: Microscopic image of the 24-Well plate with only water and no cells. The wells are visibly empty as expected. The use of this test was to ensure that the vibration was being properly coupled to the bioreactor and thus to the 24-Well plates using the vibration meter.

Objective: Run through the full motion profile with water to make sure the parafilm seals the 24-Well Plate properly and that the testing setup is transferring the vibration to the wells. If vibration is not transferring vibration, iterate on setup and repeat this test until vibration is coupled well (vibration of $\sim 30 \text{ mm/s}^2$)

Setup: The 24-Well Plate was filled with water before performing the Overall Procedure. We did not visualize the disrupted and dislodged cells as there were none in this Control Test. We did

use the vibration meter to measure vibration on the bioreactor and on the 24-Well Plate during Step 6 to make sure that the setup was coupling vibration properly.

Results: We noticed when measuring the vibration from the Netter Vibrator was not coupling well to the 24-Well Plate with a vibration of approximately $6\text{-}7\text{ mm/s}^2$ which was much smaller than expected ($30\text{-}40\text{ mm/s}^2$). We found that the extra command strip we had used to create a tight fit between the vibration plate and the right fixture was tightly pressing the vibrator against the bioreactor. When this command strip was removed, the vibration platen had a slightly loose vertical fit in the right fixture. This allowed for a small amount of space between the vibrator and the bioreactor for the vibrator to move up and down and essentially strike the bioreactor repeatedly. The change provided a much better vibration to the 24-Well Plate at approximately 30 mm/s^2 , which was much closer to the vibration we were expecting. The large central pin that runs through the center of the fixtures held the vibrator and fixture secure when rotating, so the slightly loose vertical fit for the vibration platen did not cause any issues in falling out of the frame during testing.

Test 2: 30 Second Cell Agitation WITHOUT Dissociation Enzyme and Treatment

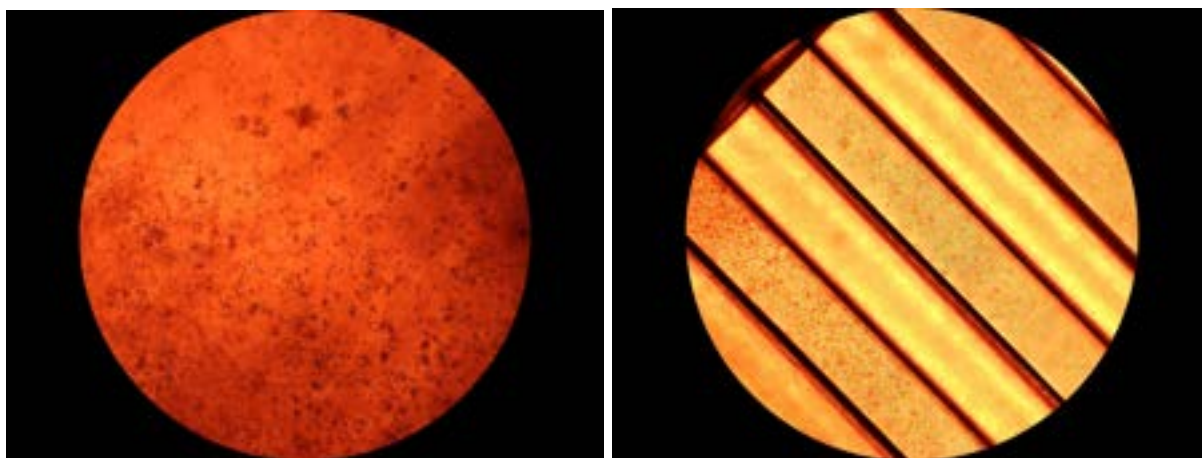


Figure 27: (Left) Microscope image of the K562 suspension cells through the top plate of the 24-Well Plate after agitation for 30 seconds. As expected, the majority of the suspension cells fell out from the wells without any dissociation enzyme and treatment. (Right) Microscope image of the K562 suspension cells through the bottom of the 24-Well Plate after agitation for 30 seconds. We can still see some of the suspension cells in the wells.

Objective: Run through the Overall Procedure with a vibration time of 30 seconds. See the effects on suspension, spheroid, and adherent cells and with NO accutase dissociation enzyme for dislodging adherent cells. Should expect the suspension cells to be majority dislodged, some of the spheroid cells to dislodge, and none of the adherent cells to drop (unless they did not already adhere to the silicon).

Setup: The 24-Well Plate was filled with media and cells and stored in the incubator before our arrival by Austin Santiago. When starting the test, the 24-Well Plate with cells was removed from the incubator and the Overall Procedure was performed. For Step 8, we had Austin check the 24-Well Plates and provide a rough estimate of the percentage of cells dropped for each type of cell as well as take images on the XDemics microscope. It should be noted that we did not measure the vibration during Step 6 but determined the vibration to be strong enough to continue the test.

Results: After 30 seconds of vibration, greater than 80% of the suspension cells dropped, approximately 10-20% of the spheroid cells dropped, and 0% of the adherent cells dropped. This matches what we expected. The test shows that 30 seconds was able to drop a substantial amount of the suspension cells without any extra treatment and that the XDemics coating to tightly adhere the adherent cells to the silicon is working very well.

Test 3: 5 Minute Cell Agitation WITHOUT Dissociation Enzyme and Treatment

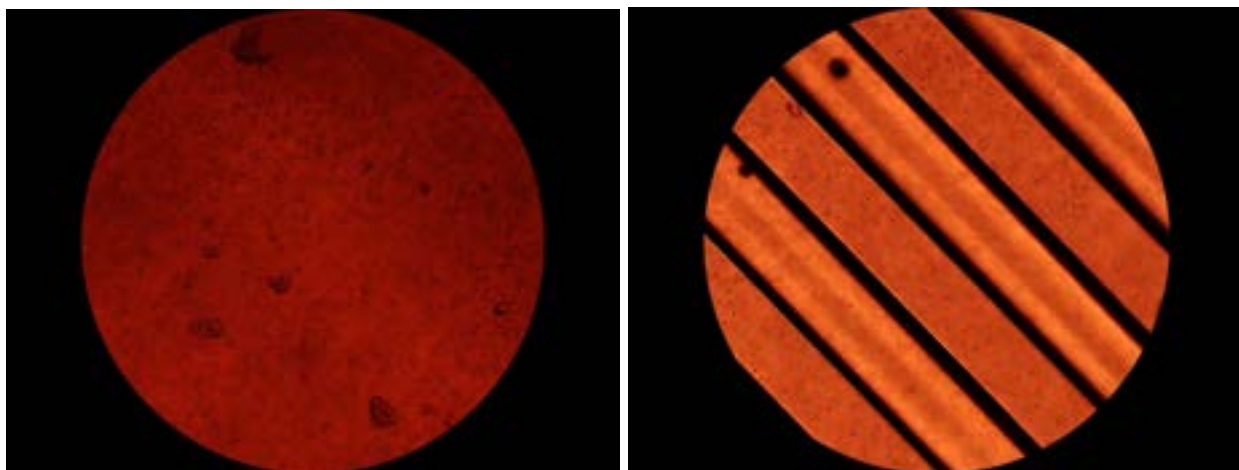


Figure 28: (Left) Microscope image of the K562 suspension cells through the top plate of the 24-Well Plate after agitation for 5 minutes. Even more of the suspension cells were dislodged from the wells. (Right) Microscope image of the K562 suspension cells through the bottom of the 24-Well Plate after agitation for 5 minutes. <5% of the cells can be seen left in the wells.

Objective: Run through the Overall Procedure with a vibration time of 5 minutes. See the effects on suspension, spheroid, and adherent cells and with NO accutase dissociation enzyme for dislodging adherent cells. Should expect nearly all the suspension cells to be dislodged, most of the spheroid cells to dislodge, and none of the adherent cells to drop (unless they did not already adhere to the silicon).

Setup: The 24-Well Plate was filled with media and cells and stored in the incubator before our arrival by Austin Santiago. For Step 8, we had Austin check the 24-Well Plates and provide a rough estimate of the percentage of cells dropped for each type of cell as well as take images on the XDemics microscope. It should be noted that we used the same 24-Well Plate as the plate from Test 2 that had been agitated for 30 seconds, went through the testing procedure and agitated for 4 minutes and 30 seconds. We also accidentally rotated the bioreactor 180° after agitation such that the wells were no longer upside down. To offset the time the cells had to resettle into the grooves, we repeated the vibration upside down for another 30 seconds before removing the plates upside down.

Results: After 5 minutes of vibration, greater than 95% of the suspension cells dropped, approximately 80% of the spheroid cells dropped, and 0% of the adherent cells dropped. This matches what we expected. The test shows that 5 minutes was able to drop the same amount of the suspension cells as the original XDemics procedure without any extra treatment and that the XDemics coating to tightly adhere the adherent cells to the silicon is working very well.

Test 4: 30 Second Cell Agitation WITH Dissociation Enzyme and Treatment

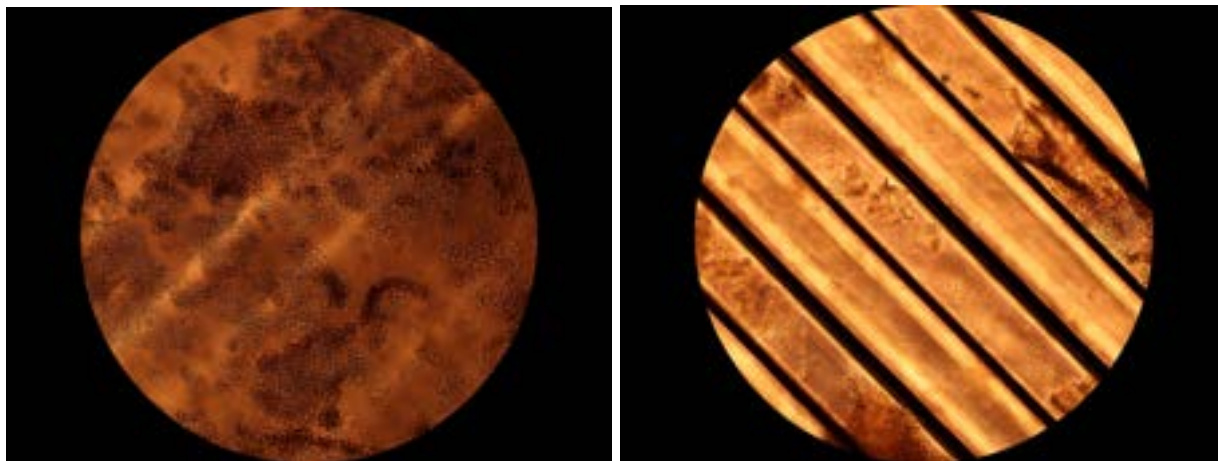


Figure 29: (Left) Microscope image of the Hek293 adherent cells through the top plate of the 24-Well Plate after agitation for 30 seconds. As expected, the majority of the adherent cells fell out from the wells when the accutase enzyme was added. No single cell suspension, mostly aggregates of adherent cells were dislodged. (Right) Microscope image of the Hek293 adherent cells through the bottom of the 24-Well Plate after agitation for 30 seconds. Portions of the cells can be seen aggregated meaning they are detached but may have settled into the well when imaging through the bottom of the wells.

Objective: Run through the Overall Procedure with a vibration time of 30 seconds. See the effects on adherent cells and WITH accutase dissociation enzyme for dislodging adherent cells. Should expect nearly all the suspension cells to be dislodged, most of the spheroid cells to dislodge, and some of the adherent cells to drop.

Setup: The 24-Well Plate was prepared on the spot by Austin Santiago. The process involved filling the plate of cells with media, removing the media, washing the cells with Phosphate-Buffered Solution (PBS), adding accutase dissociation enzyme, and then incubating the plate for 10 minutes at 37°C. When starting the test, the 24-Well Plate with cells was removed from the incubator and the Overall Procedure was performed. For Step 8, we had Austin check the 24-Well Plates and provide a rough estimate of the percentage of cells dropped for each type of cell as well as take images on the XDemics microscope. This plate was

discarded after the test and not used for Test 5. Note that we were only focusing on dropping the adherent cells so we did not get a rough estimate of drop for the spheroids or suspension cells.

Results: After 30 seconds of vibration, greater than 80% of the adherent cells dropped. This matches what we expected. The test shows that 30 seconds of vibration is able to drop a majority of the adherent cells when the accutase enzyme is included. Although this is promising our goal is >85% of the adherent cells with a stretch goal of single cell suspension.

Test 5: 5 Minute Cell Agitation WITH Dissociation Enzyme and Treatment

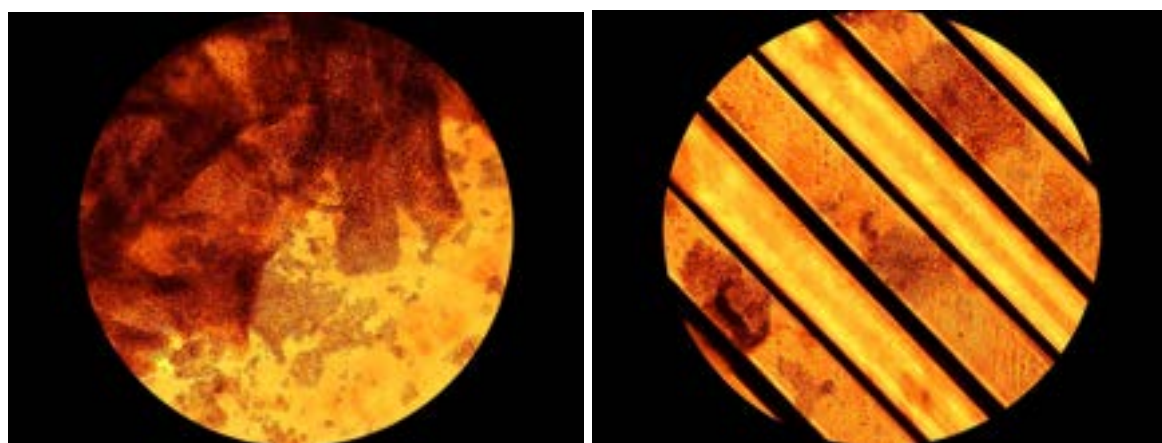


Figure 30: (Left) Microscope image of the Hek293 adherent cells through the top plate of the 24-Well Plate after agitation for 5 minutes. Even more of the adherent cells fell out from the wells when the accutase enzyme is added and the wells are put under a longer agitation time. No single cell suspension seen mostly aggregates of adherent cells, were dislodged. (Right) Microscope image of the Hek293 adherent cells through the bottom of the 24-Well Plate after agitation for 5 minutes. As before, portions of the cells can be seen aggregated and detached from the well but may have settled into the well when imaging through the bottom of the wells.

Objective: Run through the Overall Procedure with a vibration time of 5 minutes. See the effects on nonadherent and adherent cells and WITH dissociation enzyme for dislodging adherent cells. Should expect nearly all the suspension cells to be dislodged, most of the spheroid cells to dislodge, and majority of the adherent cells to drop.

Setup: The 24-Well Plate was prepared on the spot by Austin Santiago using the same procedure as Test 4. When starting the test, the 24-Well Plate with cells was removed from the incubator

and the Overall Procedure was performed. For Step 8, we had Austin check the 24-Well Plates and provide a rough estimate of the percentage of cells dropped for each type of cell as well as take images on the XDemics microscope. This plate was discarded after the test and not used for Test 6.

Results: After 5 minutes of vibration, greater than 95% of the adherent cells dropped and ~100% of the nonadherent cells dropped. This was even better than what we expected! The test shows that 5 minutes of vibration is extremely effective in dislodging the adherent cells when the accutase enzyme is included and still extremely effective at dislodging the nonadherent cells (spheroid and suspension). The adherent cells were still dislodged as clumps and not in single cell suspension.

Test 6: Intermittent Incubation WITH Dissociation Enzyme and Treatment

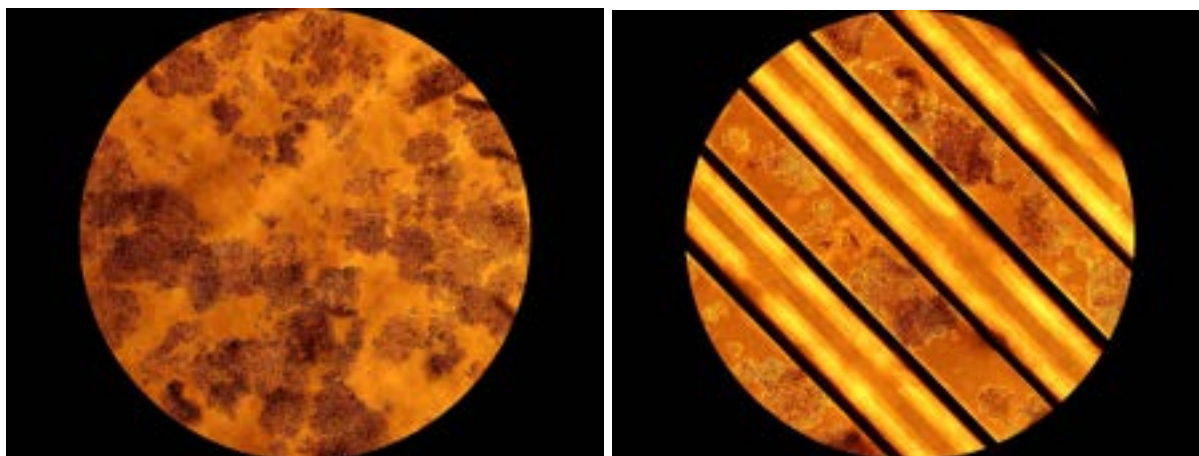


Figure 31: (Left) Microscope image of the Hek293 adherent cells through the top plate of the 24-Well Plate after alternating incubation and 30 second agitation twice. About the same amount of cells dislodged as the agitation for 5 minutes with no intermittent agitation. No single cell suspension seen mostly aggregates of adherent cells were dislodged. (Right) Microscope image of the Hek293 adherent cells through the bottom of the 24-Well Plate after alternating incubation and 30 second agitation twice. As before, portions of the cells can be seen aggregated and detached from the well but may have settled into the well when imaging through the bottom of the wells.

Objective: Run through a modified procedure that better mimics the current manual procedure where the cells are incubated, shaken, incubated, and shaken again. See if this new procedure can create single cell suspension.

Setup: The 24-Well Plate was prepared on the spot by Austin Santiago using the same procedure as Test 4 and 5 except with 4.5 mL of accutase. When starting the test, the 24-Well Plate with cells was removed from the incubator and the Overall Procedure was performed with a vibration time of 30 seconds, STOPPING after Step 7, incubating the wells upside down for another 10 minutes, and then repeating steps 4-7 with a vibration time of 30 seconds before visualizing the cells. For Step 8, we had Austin check the 24-Well Plates and provide a rough estimate of the percentage of cells dropped for each type of cell as well as take images on the XDemics microscope. This plate was discarded after the test. It is important to note that due to the large concentration of accutase which can kill cells if left too long, the procedure needs to be completed as quickly as possible.

Results: After the two sets of 30 second vibration with 10 minutes of incubation in between, the same percentage of adherent and nonadherent cells dropped as in Test 5. This time we were able to do a cell count to test viability which was around 84% which is pretty good considering that the 24-Well Plates went through a long handling time. The adherent cells were still dislodged as clumps and not in single cell suspension. However, pipetting the wells after the second vibration worked really well in separating the adherent cells after dislodgement. We believe that when the adherent cells are pushed through the tubing when exiting the bioreactor, the pressure will be able to separate the adherent cells for single cell suspension.

Overall Results

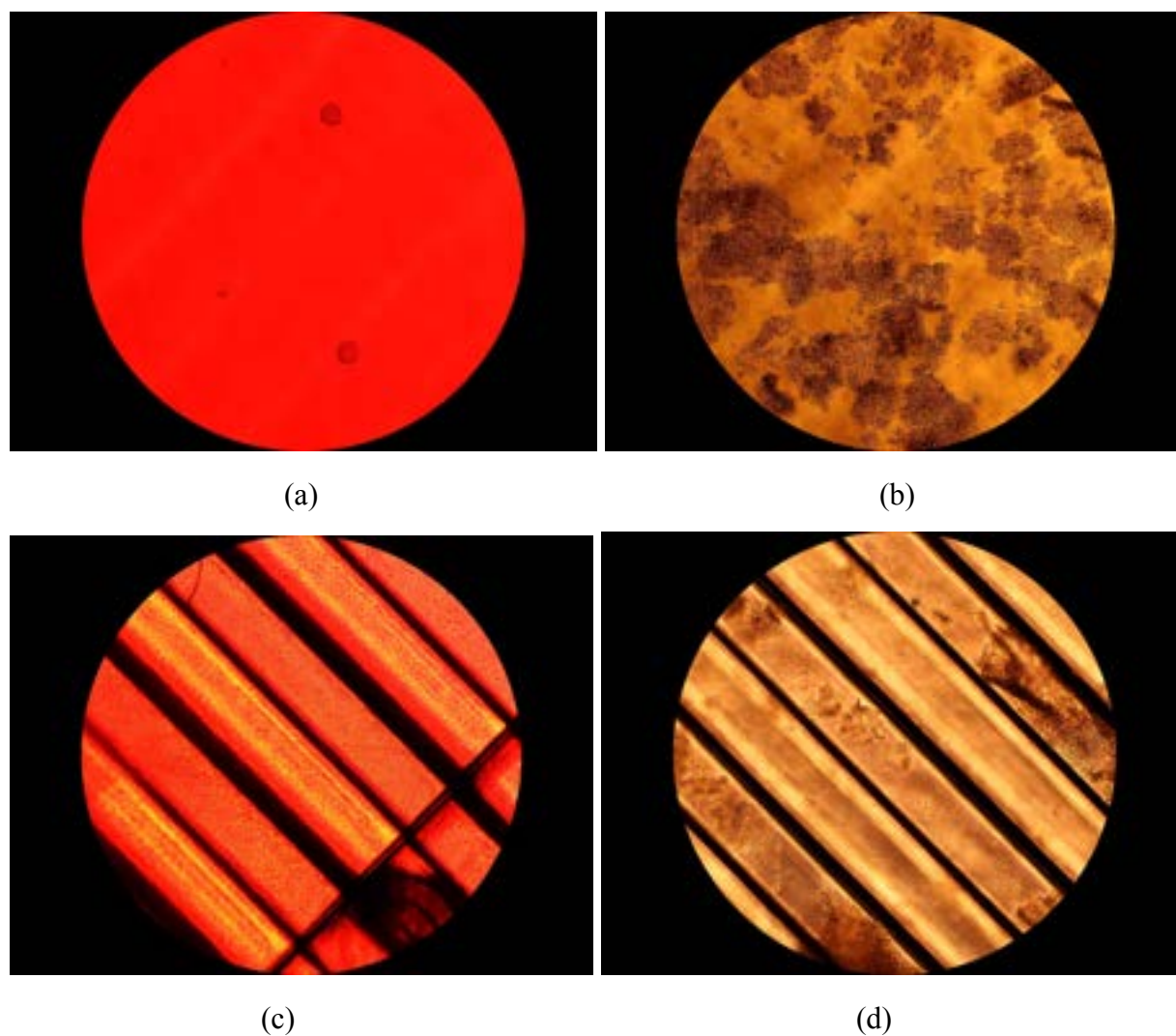


Figure 32: (a) Microscope image of the Hek293 adherent cells through the top plate of the 24-Well Plate after 30 second of agitation without the accutase dissociation enzyme where only the adherent cells that failed to adhere were visualized vs. (b) Microscope image of the Hek293 adherent cells through the top plate of the 24-Well Plate after 30 seconds of agitation with the accutase dissociation enzyme majority of the adherent cells dislodged in clumps. (c) Microscope image of the Hek293 adherent cells through the bottom of the 24-Well Plate after 30 second of agitation without the accutase dissociation enzyme where the adherent cells are densely populated and stuck to the wells vs. (d) Microscope image of the Hek293 adherent cells through the bottom of the 24-Well Plate after 30 seconds of agitation with the accutase dissociation

enzyme majority of the adherent cells dislodged in clumps and well shape can be seen meaning adherent cells not packed into the wells as in (c).

The prototype was successful at coupling the vibration from the Netter Vibrator to the 24-Well Plates attached to the bioreactor and was able to reach the same percentage of cell dislodgement as the XDemics original procedure for suspension, spheroid, and adherent cells. Although we didn't reach our stretch goal of single cell suspension for the adherent cells, it is likely that the removal of the cells from the bioreactor through the tubing will disrupt the adherent cell clumps to separate them into single cells like through light pressurization like the pipette.

Vibration Distribution

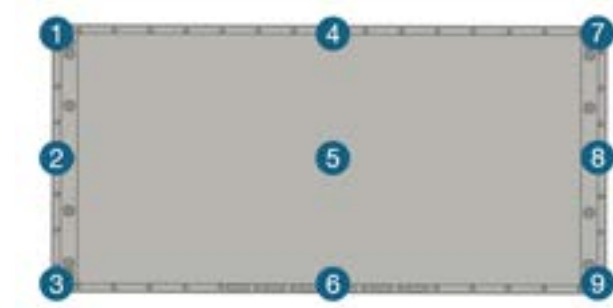


Figure 33: 9-point array of the underside of the bioreactor.

Number	Acceleration [m/s ²]	Velocity [mm/s]	Displacement [mm]
1	21	26	0.21
2	28	38	0.31
3	28	38	0.30
4	44	24	0.12
5	31	18	0.10
6	27	19	0.13
7	46	70	0.60
8	53	74	0.62

9	65	65	0.50
Mean	38.11	41.33	0.32
Standard Deviation	14.63	22.50	0.21

Table 7: Acceleration, velocity, and displacement of the vibrations measured on the underside of the bioreactor at 9 points.

The vibration transfer was measured on the underside of the bioreactor, when the vibrator was sitting on top of the bioreactor and turned on. This test was set up using the final prototype, imaged in Figure 14. The acceleration, velocities, and displacements of points 1-6 are mostly 1 standard deviation below the points 7-9. In this setup, the fixture backing was on the side of 7-9. It seems that this caused a significantly higher vibration transfer.

	Acceleration [m/s²]	Velocity [m/s]	Displacement [mm]
Not held down by hand	6.6	5.6	0.063
Held down by hand at high force	6.9	13.4	0.11
Held down by hand at the loudest vibration	52.4	43.2	0.334

Table 8: Measurements were taken using fixtures with a 1/16" gap between the vibrator and the bioreactor. Acceleration, velocity, and displacement of the vibrations measured on the underside of the bioreactor between points 2 and 5.

It was also found that the vibration transfer was very dependent on the distance between the vibrator and the bioreactor surface. Table 8 shows vibration measurements taken in a prototype setup using a previous iteration of the fixture, which had a 1/16" gap between the vibrator and the bioreactor. The vibrator was pushed down by hand, with a team member's hand on the underside of the vibrator attachment. In the third setup, the vibrator was pushed down with the amount of force that caused the loudest noise from the vibration. We found that there

was an ideal distance between the vibrator and the bioreactor where the vibration transfer was optimized. Future work could be done on finding the optimal distance for a desired level of vibration transfer. The success at the Buck Institute test shows that very little vibration transfer (6-7 mm/s²) is actually needed to dislodge cells at the XDemics specification, so maximizing the vibration transfer is not actually required.

Microsphere Testing

Following the successful validation of the wetting procedure, microsphere testing was conducted to evaluate the system's ability to introduce and evenly distribute particles throughout the HDCR. This step serves as a proxy for cell seeding performance, as uniform distribution of microcarriers is critical for achieving consistent cell attachment and growth across all membrane surfaces.

For this testing, 1 g of Cytodex microcarriers was prepared by first swelling in PBS and subsequently staining with trypan blue to allow for clear visualization during flow. Prior to introducing the microspheres, the HDCR was wetted using the previously established 40% ethanol wetting protocol to ensure all internal surfaces were fully primed and free of trapped air. Once wetting was confirmed, the HDCR was oriented vertically (90°) to allow the microsphere suspension to be introduced from the top and flow downward through the system under gravity-assisted perfusion.



Figure 34: Uneven distribution of microspheres within the HDCR showing concentration along the central flow path

Despite successful introduction of the microspheres into the system, distribution was observed to be uneven across the HDCR. As shown in Figure X, microspheres were

predominantly concentrated along the central flow path, with minimal distribution observed in adjacent channels. This indicates that the microspheres were not adequately dispersed throughout the reactor volume. The uneven distribution is likely due to a combination of tubing and pressurization inconsistencies within the perfusion system. Modifications to the tubing, including manual cutting and reattachment, may have introduced variations in flow resistance, leading to preferential flow through the central line. Additionally, partial obstruction or accumulation of microspheres within certain tubing segments may have further restricted flow to peripheral channels.

These results suggest that while the system is capable of introducing microspheres into the HDCR, further refinement of the flow path design and pressure distribution is required to achieve uniform dispersion. Future testing will focus on improving tubing consistency, minimizing flow restrictions, and optimizing system pressurization to ensure even distribution of microcarriers during seeding.

Management Plans

The Workbreakdown Structure and the Gantt Chart (see Figures 21 and 22) were used for managing the work schedule and the direction of the project. The Work Breakdown Structure was used for a more overarching look at what deliverables and key action items should be tackled each month, whereas the Gantt Chart was for setting more specific dates and timelines for working on sections of the project, as well as for assigning responsibility of the project tasks to specific team members.

Work Breakdown Structure ([link](#))

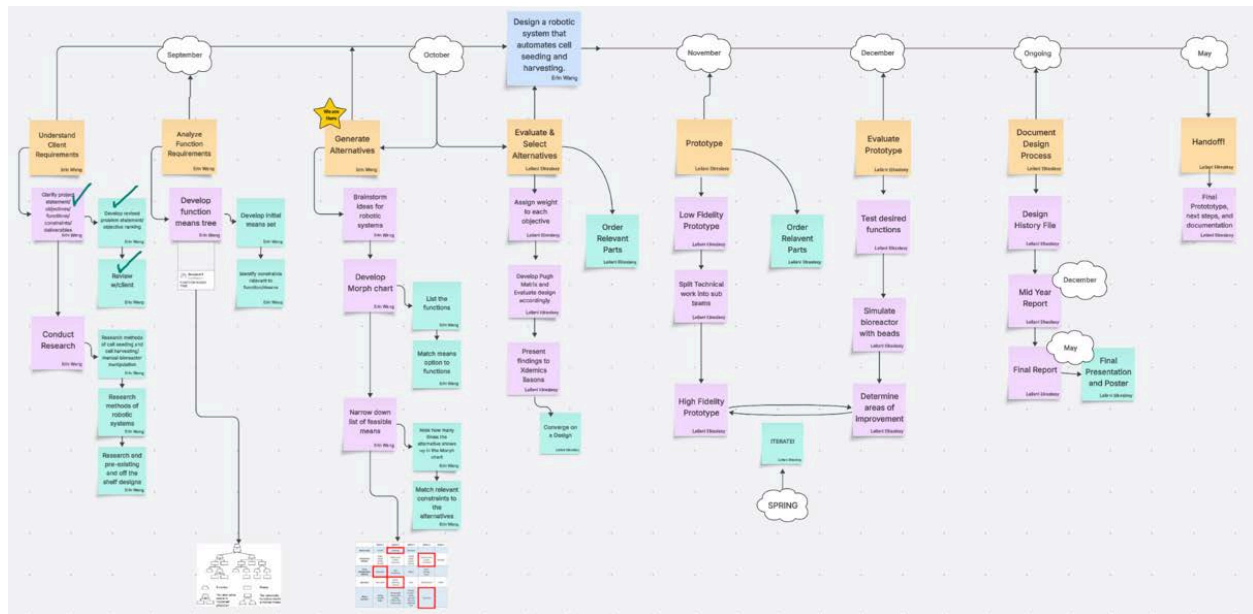


Fig 35: The Work Breakdown Structure (see Appendix D for full size version) delineates the projected work timeline, including main action items and key deliverables. In conjunction with the Gantt Chart, the Work Breakdown Structure gives an overarching view of where in the project the team should be each month.

Gantt Chart ([link](#))

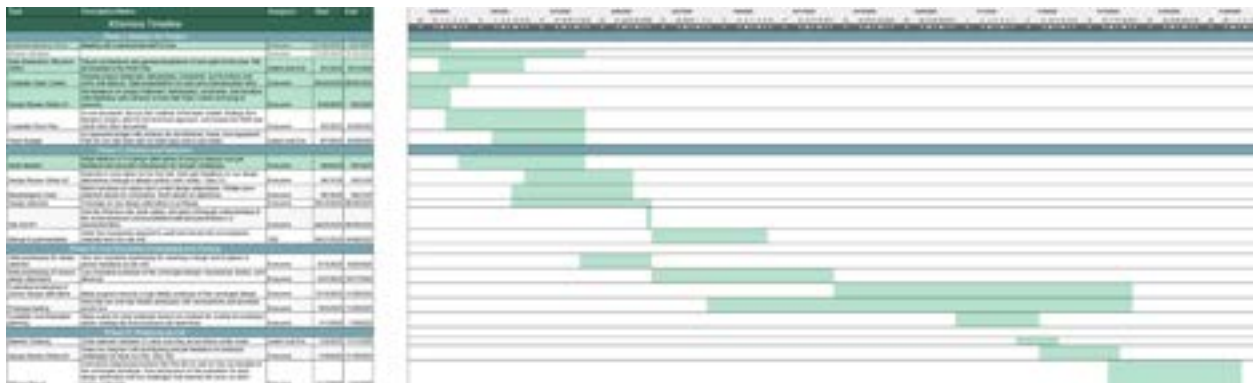


Figure 36: The Gantt Chart (see Appendix E for full size version) sets out a Fall semester schedule for key sections of the project to be completed by. In conjunction with the Work Breakdown Structure, the Gantt Chart tracks the team's progress and completion.

Task	March				April				
	WK1	WK2	Break	WK4	WK1	WK2	WK3	WK4	WK5
Wetting Testing		✓							
Vibration Testing		✓							
Finish Presentation		✓							
Practice Presentation		✓							
Determine Site Visit Goals		✓							
Prepare Site Visit				✓					
Fix Vibration Attachment				✓					
Practice (Monday)				✓					
Continue Testing				✓					
Low-Res Tilt for Site Visit					✓				
Prep for Site Visit					✓				
Site Visit					✓				
Review Site Visit Notes						✓			
Clean Up Prototype						✓			
Tilt Mechanism Iteration						✓			
Start Construction						✓			
Start Motor						✓			
Finish Poster Draft							✓		
Finish Tilt Plate							✓		
Finish Cleaning Prototype							✓		
Finish Motor							✓		
Poster Due								✓	
Final Report Draft								✓	
Revisit rendering								✓	
Final Report Due									✓

Figure 37: The Gantt Chart (see Appendix E for full size version) sets out a Spring semester schedule for key sections of the project to be completed by.

Future Work

This section outlines potential areas for future development to further improve the performance, scalability, and robustness of the automated HDCR system. Key areas of focus include manufacturing considerations, vibration optimization, fixture design, platen refinement, control system enhancements, and additional testing and validation.

Fixture Design

Future work on the fixture system should focus on improving ease of use, vibration consistency, and scalability across different bioreactor sizes. While the current fixture design

successfully secured the bioreactor and enabled effective vibration transfer, additional iterations could further optimize the geometry and tolerancing of the slots to improve alignment and repeatability. In particular, future work should investigate how vibration transfer changes across varying membrane stack heights and bioreactor thicknesses.

Additional refinements could also improve usability for cleanroom technicians by simplifying installation and removal of the bioreactor while wearing gloves. Prototype hardware such as bolts and wing nuts should be replaced with more robust quick-release mechanisms such as clevis pins or integrated latching systems. Long-term testing should also be conducted to evaluate fatigue and wear on the fixture components under repeated vibration loading to ensure durability throughout extended operation cycles.

Platen Refinement

Future work for the platen design should focus on XDemics refining and manufacturing platens with integrated extrusion features specifically designed for the fixture system developed in this project. The current prototype demonstrated that platen extrusions provide a simple and effective method for aligning and securing the bioreactor within the fixture while also improving repeatability during setup and operation. Future iterations of the platens should therefore incorporate these extrusion features directly into the manufactured platen geometry rather than relying on prototype modifications or add-on components. Additional work should also focus on optimizing the dimensions, tolerancing, and placement of the extrusions to ensure reliable engagement with the fixture system across different membrane stack heights and bioreactor configurations.

Vibration System

Future work on the vibration system should focus on improving the consistency and controllability of vibration transfer throughout the bioreactor. The vibration distribution measurements showed that points near the fixture backing experienced significantly higher acceleration, velocity, and displacement compared to other regions of the bioreactor. This suggests that the current fixturing geometry and contact conditions create nonuniform vibration

transfer across the system. Additional iterations of the fixture design could therefore be performed to improve vibration uniformity and reduce variability between measurement locations.

The testing also demonstrated that vibration transfer was highly dependent on the distance and contact force between the vibrator and the bioreactor surface. An optimal spacing appeared to exist where vibration transfer was maximized, indicating that tighter manufacturing tolerances or an adjustable mounting mechanism could improve repeatability of the system. Future work could involve systematically characterizing the relationship between gap distance, applied force, and resulting vibration transfer in order to identify a target operating range for reliable performance.

However, results from testing at the Buck Institute demonstrated that only a relatively small amount of vibration transfer was required to successfully dislodge cells according to XDemics specifications. Because of this, future development should prioritize achieving consistent and repeatable vibration transfer rather than maximizing vibration amplitude. Additional testing could also be performed to determine the minimum vibration thresholds necessary for effective cell removal, allowing the system to be optimized for reliability while minimizing unnecessary vibration and mechanical stress on the bioreactor.

Control System

The proposed control system utilizes a 3 N·m closed-loop NEMA stepper motor with an integrated electromagnetic brake, coupled to the bioreactor through a timing pulley system to enable controlled rotation. This configuration was selected to provide sufficient torque, precise positional control, and ensure a safe system state is reached in the event of power loss. While the components of the potential control system have been analyzed and designed based on its performance characteristics, no further integration has been done as of now. As such, the current system does not yet include automated rotational control.

Future work should consider the full integration of the control system into the prototype. This involves writing software capable of generating step pulses and directional signals required for motor control, then using the information provided by the closed loop system for accurate control and monitoring of the motor and HDCR. Further considerations for development would

involve the programming and implementation of the electromagnetic braking system as well as the implementation of a user-friendly interface that enables easy interaction between the operator and the device.

Manufacturing Considerations

Future work for the final manufacturing of the vibration and spinner system should focus on transitioning the prototype into a fully cleanroom-compatible and manufacturable design. Components that directly interface with the bioreactor or operate within the cleanroom environment should be manufactured from materials such as stainless steel to minimize particle generation, improve durability, and ensure compatibility with cleaning chemicals and sterilization procedures. The fixture system was designed with manufacturability in mind, including simplified geometries, scalable slot features, and tolerancing intended for repeatable fabrication and assembly.

Additionally, several prototype components can be replaced with commercially available hardware in a final manufactured system to improve reliability and simplify assembly. Structural elements such as the frame could be manufactured from welded or machined steel rather than 80/20 extrusion, while bearings, shafts, hinges, motors, and fasteners could be sourced as off-the-shelf industrial components. Future iterations should also incorporate tighter manufacturing tolerances and refined mounting methods to improve repeatability of vibration transfer and overall system robustness for long-term operation in a laboratory environment.

Future Testing

One of the stretch goals that the team didn't get to this year was optimizing the motion profile of the bioreactor to maximize cell throughput. Now that we have created a system able to achieve the necessary wetting angle and full rotation along the short axis of the bioreactor, we would work towards achieving this stretch goal by modularly testing the rotation angle of the bioreactor when seeding, growing, harvesting, agitating, and transfecting and measuring the cell throughput of the system to find the best motion profile for the cellular process.

Due to timing constraint and damage to the bioreactor seal during transportation from the Site Visit back to Harvey Mudd, we also weren't able to accurately simulate what the distribution of cells would be in the bioreactor using our spinner mechanism. That means our tests on cells were only captured on the 24-Well plate and not in the 5-stack bioreactor. Thus for future testing, we would use a new, completely sealed bioreactor and profuse the system with cells to observe whether our system promotes even distribution of the cells amongst the wells.

A final future test would be to optimize the agitation frequency provided by the Netter vibrator. We only had time to work at the Netter's default frequency of 50 Hz, so future testing would allow us to manipulate the weights inside the vibrator to produce different frequencies of vibration and measure the effects of varying frequencies in dislodging the cells.

Appendix A: User Design Requirements

Critical user Requirement (nice to have)	Design Input (Specification)
Must be able to manipulate profusion reactor	Device supports full movement envelope for a 50 stack tall device for a wet and dry mass, including eccentric fluid mass. Justification: bioreactor will need to be manipulated with varying amounts of liquid and masses inside the stack as well
Must support wetting	System removes ≥ 90 percent of bubbles during wetting/agitation phase. Justification: Remaining bubbles inside the silicon grooves will prevent cells from settling into the grooves and greatly decrease cell throughput
Must support failures (e.g. power outage, system glitch)	Automated safe state and resume procedure. System locks in place when event occurs and logs the failure. Justification: A loss of power could trigger electrical components to fail and result in loss of stability and control
Usable while wearing standard PPE (e.g. no screws)	No screws or small tools required for handling system in the clean lab. Handles and latches operable with lab gloves 1 size larger than optimal fit. Justification: In the clean lab there cannot be any small components and users are wearing gloves which reduces handling dexterity
Must dislodge cells from wells effectively	System supports upside-down rotation and agitation in the upside-down position at a define amplitude/frequency. Justification: An upside-down position inverts the wells to allow for cells to move away and out from the silicon grooves. Agitation will likely be necessary to completely dislodge cells from wells.
Must not damage cells	Shear stress on cells of applied to cells < TBD. Maximum agitation and rotation acceleration < TBD. Justification: Quick or harsh movements could exert a force that damages the cells.
Keep cells in suitable conditions during growth phase	Bioreactor tilt angle must remain at $5^\circ \pm 1$ TBD for $\geq$ TBD hours. Justification: If kept flat, the bioreactor accumulates condensate that disrupts the distribution and settling of cells.
Simple user setup in lab environment	System has < 5 major steps to start. The fixtures for inserting bioreactor into system is coupled. User interface with system controls and procedures. does not require programming experience. Justification: The users are biologists in labs with gloves on. Keeping the setup simple and easy to use is crucial for the system to be beneficial over manual manipulation.
Must not add pressure to top and bottom of bioreactor	No fixtures applying unequal or compressive loads on top and bottom of bioreactor. No use of clamps on the top and bottom of bioreactor. Justification: A compressive force or unequal forces on the top and bottom of the bioreactor could result in unequal spacing and thus unequal pressure flows that will affect consistency of conditions between stacks.
Must be suitable for a clean room	All materials must be cleanable with IPA and compatible disinfectants, no crevices that cannot be reached by user with gloves and standard cleaning equipment. Justification: Because the system will be reused for multiple cycles, the system needs to be able to be fully cleaned between operation cycles with standard clean room equipment and solutions
Low-particle generating	System produces < TBD particles during operation. Justification: The system must abide by clean room standards which does not allow materials that produce substantial amount of particles.
Future to the given range of reactor sizes	Adjustable fixture supports 15 stack to 50 stack reactors without requiring replacing the system. Justification: Labs may be working with various reactor sizes for the same system and so the system fixturing to the reactor must be able to accommodate different sizes.

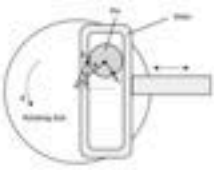
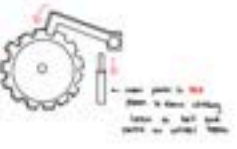
Design Outputs (Specs, drawings, BOMs, procedures)	Tests	Results	Owner
Mechanical drawings for motion profile, torque/speed specifications, load rating for motors and attachments between subsystems	Attach various masses at different positions on the bioreactor and operate through full range of motion. Verify system stability and control.	NA, testing in progress	Rotation substream/Fixturing substream
Wetting sequence procedure; Tilt angle specifications for wetting.	Visual inspection; bubbles before vs. after wetting sequence $\geq 90\%$ reduction	NA, testing in progress	Wetting Substream
Motor safety and locking specifications, Load rating of motor without power, software/firmware for logging event and system behavior	Conduct 10 power-loss test at different stages of operation; verify safe shutdown and correct lock and resume behavior	NA, testing in progress	Rotation substream/Motor substream
Assembly drawings and CAD, mechanism design, ergonomics spec	Conduct 10 test runs of full operation procedure with users wearing clean lab nitril or nitril gloves that are 1 size larger than their optimal size; measure success rate and time, verify that steps requiring interaction with system run smoothly	NA, testing in progress	Fixturing substream/UI substream
Agitation profile, motor load and control specifications, bioreactor measurements for rotation clearance	Conduct 10 test runs of cell agitation. After agitation inspect for remaining cells; quantify cell detachment; verify system has not failed nor deformed and can continue multiple agitation cycles without wear	NA, testing in progress	Agitation and experimentation substream/Rotation substream
Motion profile spec, computational fluid simulation on COMSOL, manual procedure acceleration/speed specifications	Conduct 10 test runs of full operation procedure. Before and after each agitation and each rotation, visual inspection (and measurements if possible) of current density of viable cells.	NA, testing in progress	Fluids substream/Rotation substream
Manual procedure time specifications for each step	Monitor tilt angle stability over TBD hours; verify cell health after growth phase.	NA, testing in progress	Rotation substream/Fixturing substream
User workflow instructions, user interface design (software/mechanical), fixture design, locking mechanism designs	User testing survey; timing measurements for setup, qualitative observations of setup for user with gloves on	NA, testing in progress	Fixturing substream
Fixture design, platen design, material choice, clearance dimensions, bioreactor dimensions	Visual inspection of points of contact between fixture and bioreactor, measure compressive forces	NA, testing in progress	Fixturing substream
Material specifications, cleaning procedure	Wipe-down tests; inspection for residues	NA, testing in progress	All substreams
Material selection, wear analysis, clean room particle specifications	Particle counter test after operation cycle. Particle counter test after multiple operation cycles. Wear inspection of system after multiple cycles.	NA, testing in progress	All substreams
Fixture drawing and dimension verification, fixture design, bioreactor dimensions for various stack sizes	Observation and check support and operation on all expected sizes. User testing on ease of switching between stack sizes.	NA, testing in progress	Fixture substream

Appendix B: Failure Mode and Effective Analysis (FMEA)

Process Step / Item	Potential Failure Mode	Potential Effects of Failure	Severity (5)	Potential Causes of Failure	Occurrence (0)	Current Controls	Detection (10)	RPN (S-O-D)	Recommended Actions
(Describe the step or item being analyzed)	(What could go wrong?)	(What happens if failure occurs?)	(1-10)	(Why it could happen)	(1-10)	(Existing preventive/detective controls)	(1-10)	(Auto-calculated)	(Suggested mitigation actions)
Pin	Pin could break inside the pinhole	Possibly means bioreactor is stuck to fixture and need to replace whole system	10	Choose the wrong pin, or pin experiences unforeseen amount of force	1	Buy a pin rated for calculated forces or above	2	20	Choose a pin rated above the forces expected
	Pin might fall out	Bioreactor falls out of fixture and cells are now everywhere :(	10	Mechanism holding pin failed	3	Spring loaded ball that is default locked, human detection	8	240	Possibly a stop on the system that only allows the system to move if the pin is locked, add a second pin (one pin on each side); choose pin that mitigates this failure mode
	Pin might get loose	Not strongly coupled so vibrations and force transmissions are weaker	9	Vibrations might cause pin to shake loose	3	spring loaded ball, human detection	3	81	Buy vibration resistant pin, has stronger springs and more balls.
	Pinhole might break	Bioreactor falls out of fixture, or bioreactor can't be removed.	10	Combination of vibrations and forces during operation might wear the pinhole down during use.	4	Human detection	2	80	Make the pinhole out of something strong that won't break or deform like stainless steel
	Pin getting jammed in pinhole	Can't close or open fixture. Can't start or stop process.	8	Fraction? Things falling into pin hole	5	Human detection.	2	80	Make the pinhole go all the way through the part so that anything that falls in can go out the other side.
	Pin gets lost	Can't secure bioreactor.	6	Pins aren't captive	4	Lead rope tying it to the robot.	4		Make pins captive, or connected to machine by a wire. Use a latch instead?
Latch to secure fixture to bioreactor	Latch slides open during vibration	Bioreactor falls out of fixture and cells are now everywhere :(	10	Latch improperly closed. Fatigue. Latch mechanism fails, becomes more likely after the below failure	3	Human detection.	2	60	Tight-Hold draw latch uses springs to mitigate vibrations
	Latch breaks	Possibly means bioreactor is stuck to fixture and need to replace whole system	8	Choose the wrong latch, or pin experiences unforeseen amount of force	1	Human detection	2	16	Place latch such that it experiences minimum amount of forces acting against it.
Placing bioreactor into the fixture	Bioreactor gets stuck midway and can't be taken out (entrance)	Whole system needs to be replaced	8	Deformation of fixture due to temperature or stress	1	Human detection	5	40	Manufacturing the fixtures with the right tolerance grades.
	Bioreactor gets stuck midway and can be removed	Whole system needs to be replaced	7	Deformation due to temperature or stress	1	Human detection	5	35	Manufacturing the fixtures with the right tolerance grades.
	Bioreactor expands and starts bending to the fixture	Pins can pop out, bioreactor can get stuck	10	High temperatures	2	human detection	2	40	Is this a flaw with the bioreactor or the robot? Maybe potentially make the fixture adjustable size?
Tubing Clip	Clip Break	Tube tangle	6	Tubes too tight creates torque	1	Strong Clip- Stuck in Tube	8	48	Clear user instructions- Testing failure scenarios to better design improvements
	Tube Pinch	Restrict Tube Flow	9	Torque over/night clamp	3	Too little slack in tube length.		0	
	Generate Particles							0	
Tilt mechanism	Motor stops working	Tilt is fixed at certain angle	4	Loose power, motor use failure	1	Motor safemode	8	32	Integrate safemode with some alert/detection system
	Block get stuck in wedge	Tilt is fixed at certain angle and can't be removed	5	Misalignment of wedges and block, weight of bioreactor/system too heavy	1	Human detection	2	10	Ensure sturdy materials, track system to keep block in place, account for many times expected weight
	Supports fail	Entire bioreactor system falls	10	Too much weight, stress over time	1	Human detection	2	20	Ensure system can support much more than expected weight

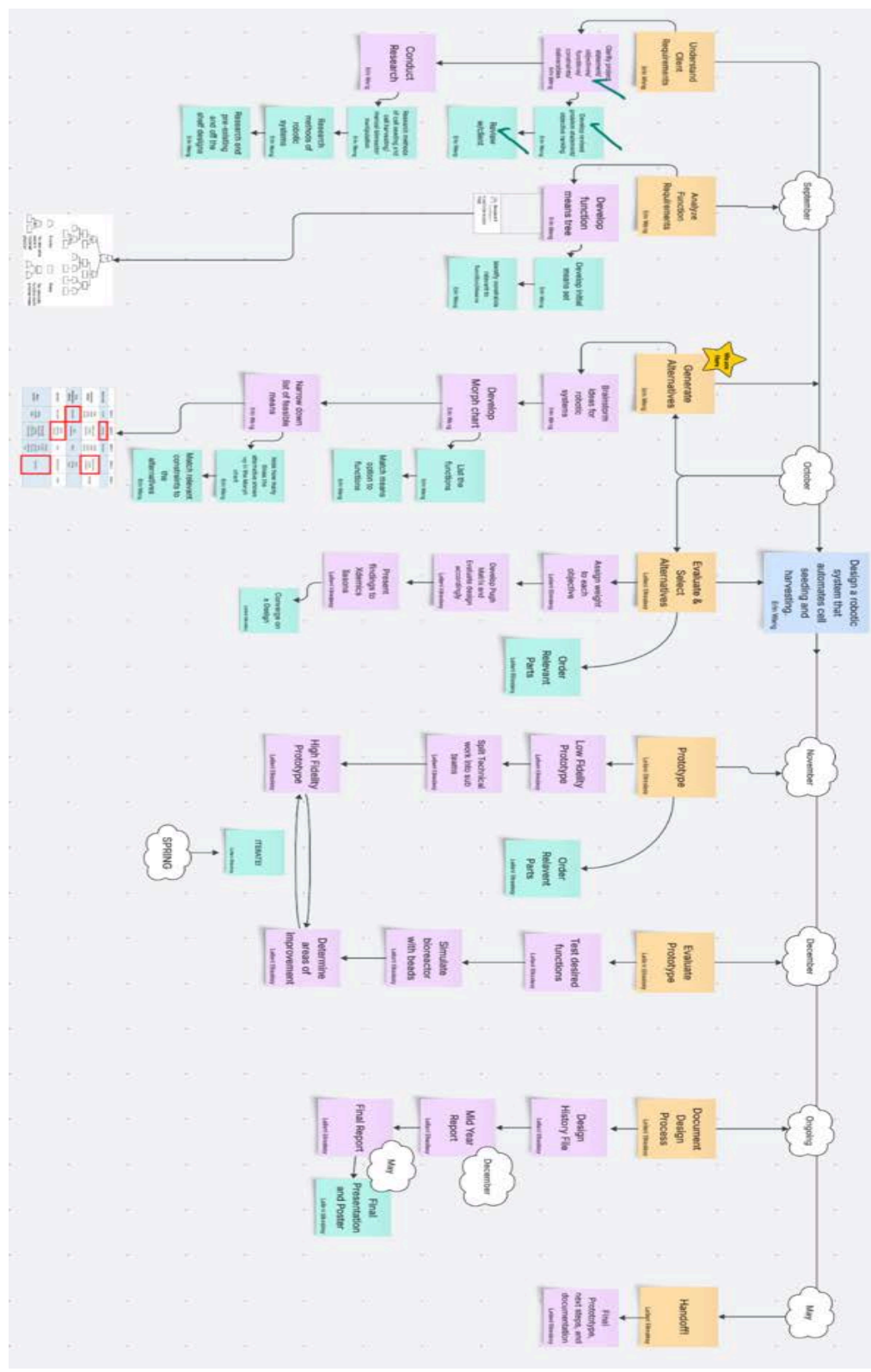
Appendix C: Morphological Chart

Functions	Means		
Rotate bioreactor along the short axis	 Tilt Plate	 Face rod and motor	 Gyroscope rings
Tilt bioreactor along the long axis	 Motor spinner	Rail system	 Gyroscope rings
Invert bioreactor (not limited to along the short axis)	 Motor spinner	 Face rod and motor (short axis)	 Gyroscope rings (short axis)

Agitate bioreactor	 ERM coin vibration	 Scotch yoke	Microfluidic Pulses
Attach to bioreactor	 Clamps	Free-spinning fixture	 2 layers of frames
Remain in set position during failure	Control system Lock	Battery Power	 Wheel Brake

Adjust to multiple bioreactor sizes	Dovetail with lock  Dovetail	 Slider	 Lunchbox
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Appendix D: Work Breakdown Structure



Appendix E: Gantt Chart Fall and Spring

Progress	Task	Description/Notes	Assignee	Start	End
XDemics Timeline					
Phase I: Restate the Project					
☒	Establish Meeting Times	Meeting with Liaisons/Internat/Prior Lee	Everyone	8/30/2025	9/2/2025
☒	Review Literature		Everyone	8/30/2025	9/18/2025
☒	Work Breakdown Structure (WBS)	Figure out timelines and general breakdown of work path for the year. Will be included in the Work Plan	Lailani and Erin	9/1/2025	9/11/2025
☒	Complete Team Charter	Restate project statement, deliverables, constraints, and functions and verify with liaisons. State expectations for work and communication ethic with Medtronic and LSPower to then edit Team Charter and bring to Liaisons.	Everyone	08/30/2025	09/05/2025
☒	Design Review Slides #1		Everyone	8/30/2025	9/2/2025
☒	Complete Work Plan	In one document, discuss the contents of the team charter, findings from literature review, plan for the technical approach, and include the WBS and Gantt chart (this document)	Everyone	9/2/2025	9/18/2025
☒	Finish Budget	An approved budget with sections for discretionary, travel, and equipment Plan for two site visits with no hotel stays and a van rental	Lailani and Erin	9/7/2025	9/18/2025
Phase II: Ideation and Selection					
☒	Initial Ideation	Initial ideation of 3-4 design alternatives to bring to liaisons and get feedback and possible workarounds for foreseen challenges.	Everyone	09/04/25	09/18/25
☒	Design Review Slides #2	Brainstorm more ideas for the first half, then gain feedback on our design alternatives through a debate activity (with candy) (Sep 23)	Everyone	09/12/25	09/23/25
☒	Morphological Chart	Match functions to means and current design alternatives. Whittle down selection based on constraints. Rank based on objectives	Everyone	09/10/25	09/23/25
☒	Design selection	Converge on one design alternative to prototype.	Everyone	09/10/2025	09/26/2025
☒	Site Visit #1	Visit the XDemics site, grab videos, and gain a thorough understanding of the whole bioreactor process/pipettas/materials/specifications of environment/etc.	Everyone	09/26/2025	09/26/2025
☒	Manual Experimentation	Verify the movements required to seed and harvest the microspheres collected from the site visit	TBD	09/27/2025	11/14/2025
Phase III: Low Resolution Prototyping and Testing					
☒	Initial prototyping for design selection	Very low resolution prototyping for selecting a design and to gather in person feedback at site visit	Everyone	9/18/2025	9/26/2025
☒	Initial prototyping of chosen design alternative	Low resolution prototype of the converged design: mechanical, fluids, and electrical	Everyone	9/27/2025	11/7/2025
☒	Expanded prototyping of chosen design alternative	Make progress towards a high fidelity prototype of the converged design	Everyone	10/18/2025	11/20/2025
☐	Prototype testing	Test both low and high fidelity prototypes with microspheres and provided acrylic box	Everyone	10/3/2025	12/5/2025
☐	Scalability and finalization planning	Make a plan for what materials need to be ordered for scaling the prototype and/or creating the final prototype with lead times	Everyone	11/1/2025	12/5/2025
Phase IV: Wrapping up Fall					
☐	Material Ordering	Order relevant materials to make sure they arrive before winter break	Lailani and Erin	11/8/2025	12/5/2025
☒	Design Review Slides #3	Share our progress with prototyping and get feedback on prototype challenges we have run into. (Nov 18)	Everyone	11/9/2025	11/18/2025
☐	Midyear Report	Summarize progress/procedure from the fall as well as how we decided on the converged prototype. Give background on the inspiration for each design alternative and the challenges that deemed the ones we didn't choose unfeasible.	Everyone	11/13/2025	12/2/2025

Task	March				April				
	WK1	WK2	Break	WK4	WK1	WK2	WK3	WK4	WK5
Wetting Testing		✓							
Vibration Testing		✓							
Finish Presentation		✓							
Practice Presentation		✓							
Determine Site Visit Goals		✓							
Prepare Site Visit				✓					
Fix Vibration Attachment				✓					
Practice (Monday)				✓					
Continue Testing				✓					
Low-Res Tilt for Site Visit					✓				
Prep for Site Visit					✓				
Site Visit					✓				
Review Site Visit Notes						✓			
Clean Up Prototype						✓			
Tilt Mechanism Iteration						✓			
Start Construction						✓			
Start Motor						✓			
Finish Poster Draft							✓		
Finish Tilt Plate							✓		
Finish Cleaning Prototype							✓		
Finish Motor							✓		
Poster Due								✓	
Final Report Draft								✓	
Revisit rendering								✓	
Final Report Due									✓

Appendix G: Early Fixture Ideation

The following articles in Appendix G were from the ideation phase during the Fall semester:

Slider with Latch

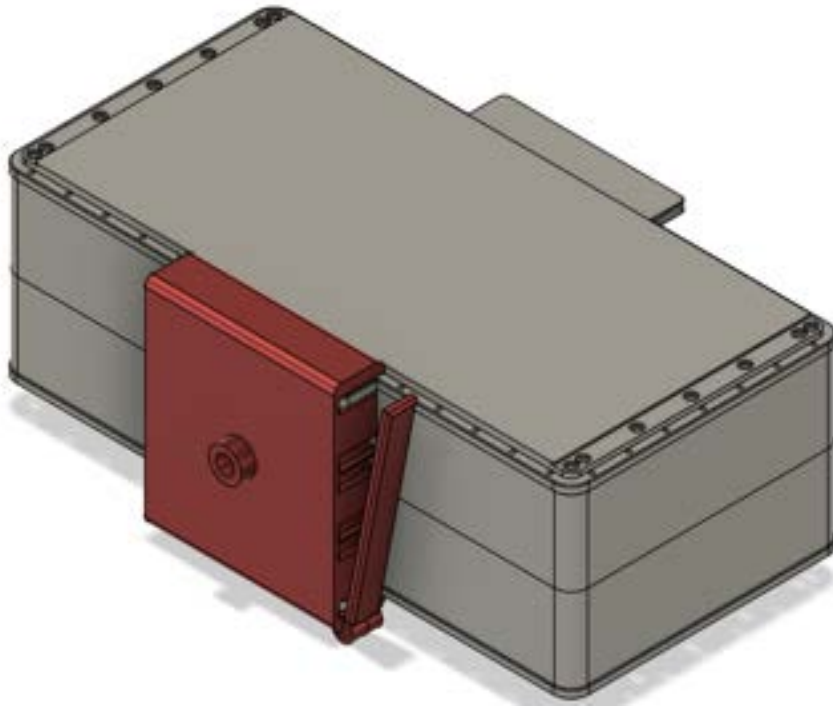


Fig 38: The Slider depicted above uses a latch instead of a pin. Using pins was ultimately chosen over using latches for the slider.

This design works identically to the Slider with a pin, except that the platen extrusions do not have through-holes for pins. Instead, the Slider fixture has a latch that closes over one side of the extrusions to constrain them from falling out when the bioreactor is rotated 90 degrees.

Between latches and pins for securing the fixture to the bioreactor, pins are advantageous because they are easier to design and manufacture to fit the Slider, and pins are simpler to work with for gloved hands. This design will not be continued, since using a pin is more desirable than a latch for the Slider.

Lunchbox

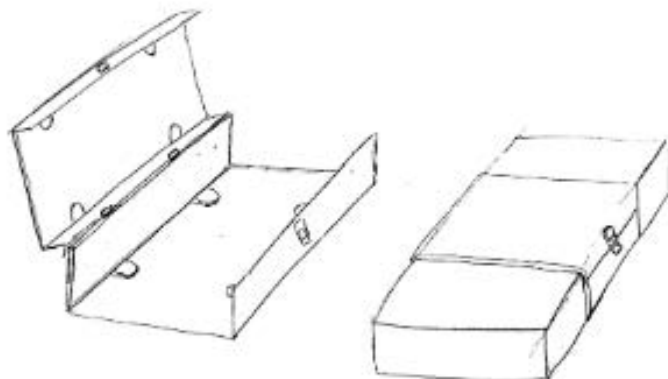


Fig. 39: The Lunchbox design alternative is depicted above. The fixture wraps around the bioreactor, which is the smaller rectangular prism in the drawing. The fixture is sealed around the bioreactor through a clasp on one side, and small extrusions on the inside of the design will fit into the grooves on the bioreactor to prevent slippage.

The Lunchbox design wraps around the bioreactor and constrains it in all directions. The half-circles on the inner faces of the Lunchbox are ridges that fit into notches cut into the top and bottom surfaces of the platens and thus constrain the bioreactor along its long axis. This prevents the bioreactor from slipping out when rotated 90 degrees. The use of only one latch makes it easy for the user to fasten the fixture around the bioreactor.

The Lunchbox's biggest disadvantage is its inability to accommodate multiple bioreactor sizes. As currently designed, the fixture would need to be switched each time a different bioreactor size was used.

Because the Lunchbox covers a majority of the Lunchbox surface area, this design has potential for integrating an incubation system into the fixture itself. However, iterating the design to include an incubation system is outside the scope of this project, and as such, is not taken into consideration when ranking the design alternatives for the fixture.

Dovetail Slider

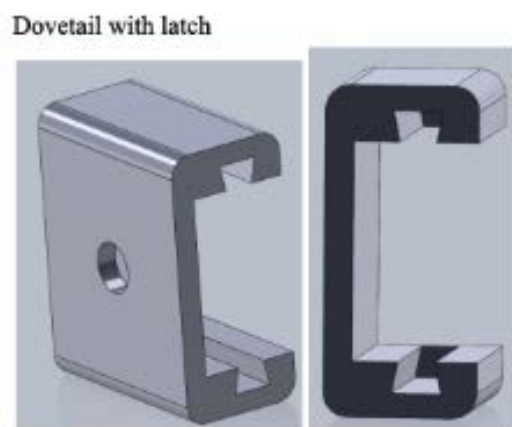


Fig.40: Depicted above is the first iteration of the Slider, which used trapezoid-shaped slots to fit trapezoid-shaped extrusions from the top and bottom surfaces of the platens.

Similar to the Slider, the Dovetail slider features extrusions on the platens that would slide into notches in the fixture. These extrusions added to the platen would protrude from the top and bottom faces of the platen instead of from the sides. The open side of the extrusions would be constrained using latches, similarly to the Slider above.

Because the platens are fabricated in a CNC milling machine, the angled dovetail extrusions would be more challenging to manufacture. This additional difficulty is unnecessary since the dovetail extrusions would constrain the bioreactor in the same way as the Slider above. Additionally, extrusions on the top and bottom faces prevent the Dovetail slider from accommodating all membrane stacks in the same way the Slider can. Compared to the Slider, the latches on the Dovetail slider would be more oddly shaped and difficult to fasten with gloves.

The only advantage of the Dovetail slider comes from the extra durability of the extrusions. The extrusion on the Slider above is limited to 6 mm, while the dovetail extrusions can be wider if necessary. However, this advantage is negligible because the platen material is Silicon Nitride, an extremely strong material.

Quick Clamp

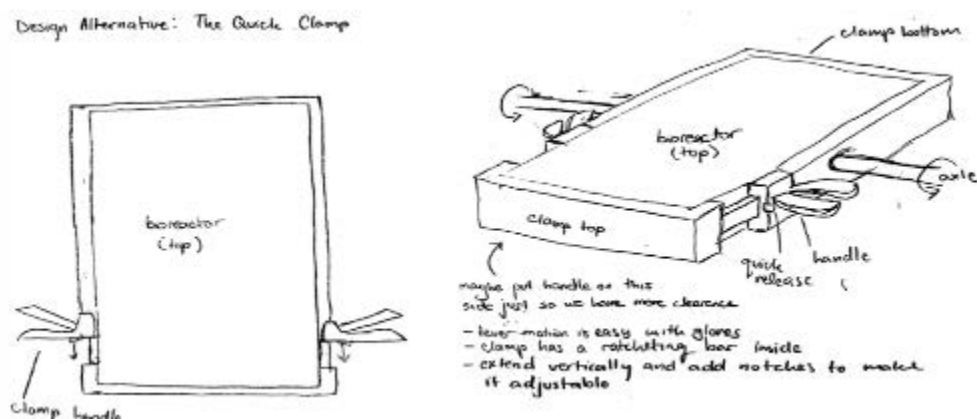


Fig. 41: The Quick Clamp design is depicted above, when it is clasped around the bioreactor.

Fixture Best of Class Chart

A Best of Class chart was created to quantitatively compare each of the fixture designs. Each design was ranked on a scale of 1-5 to how well it meets the given objective with a 5 being that it was most effective for the given objective. The objectives are ranked from most crucial to the project (7) to the least crucial (1). After ranking each fixture the quick clamp was determined to be the best design alternative.

Rank	Objective	Slider/Latch	Slider/Pin	Lunchbox	Dovetails
2	Low particle generating	4	5	3	1
5	easy to handle with gloves	2	5	4	1
6	adjustable for all stacks (5=yes, 1=no)	5	5	1	1
1	fixture is durable	1	3	2	5
3	withstand motion	1	2	3	4
4	ease of manufacturing	4	5	1	2
7	immobilize reactor in fixture	3	3	4	3
	Total	89	115	75	59

Table 9: The Fixture Best of Class Chart ranks the fixture design alternatives on a scale of 1-5 based on how effectively the design will meet an objective. A higher number on the 1-5 scale delineates a better score and a more effective design for the given objective. The objectives are also ranked from most crucial to the project (7) to least crucial (1). From the Best of Class Chart, the quick clamp was determined to be the best design alternative.

Appendix H: Manufacturing Instructions

The fixture is to be machined from a solid block of stainless steel. As it is subject to wear from repeated insertion of bioreactors and securing vibrating components, a strong material is crucial. From the stainless steel block, slots can be milled in a CNC milling machine at desired spacing appropriate with bioreactor sizing and platen thickness. Ensure a backing remains at the end of each slot to stop bioreactor insertion at proper depth. Create a vertical hole for the clevis pin all the way through the fixture.

The prototype base is assembled from aluminum 8020 extrusions. For the final product, a stainless steel frame is appropriate for cleanroom compliance, frame strength, and reliability. Polycarbonate or other plastic coverings are appropriate for aesthetics and other components such as a door to the enclosure.

The flanges on the bioreactor are to be machined from the same piece of polycarbonate the rest of the bioreactor platens are made from. Having one continuous part ensures ideal vibration transfer with no extra damping, and repeatable, precise sizing to fit into the fixture.

The vibrator platen is to be made of a cleanroom compliant, semi-flexible material. In prototyping, acrylic was used to achieve desired results. A material with similar properties that is cleanroom compliant and easy to procure for manufacturing would be appropriate.

Appendix I: Bill of Materials (BOM)

[BOM exported from CAD assembly](#)

	Part Number	Part Name	Description	Supplier	Quantity	Material	Subassembly
1		Bioreactor		In house	1	Steel	
2		8020 28"		Supplier	2	Steel	Frame
3		8020 14"		Supplier	2	Steel	Frame
4		8020 14"		Supplier	1	Steel	Frame

5		8020 18"		Supplier	2	Steel	Frame
6	2820T4_Dry-Running Mounted Sleeve Bearing	2820T4_Dry-Running Mounted Sleeve Bearing	STEP AP203	Supplier	2	Steel	Motor
7		Fixture Bridge		In house	2	Steel	Fixture
8		Fixture Final Design	Connects fixtures together	In house	2	3D printed	Fixture
9	1870K29_Easy-Access Flange-Mount Linear Shaft Support	1870K29_Easy-Access Flange-Mount Linear Shaft Support	STEP AP203	Supplier	1	Aluminum	Fixture
10		Shaft Rod		Supplier	1	Steel	Fixture
11		Clevis Pin		Supplier	2	Steel	Fixture
12	NEMA stepper motor	Motor		Supplier	1	Discrete Component	Motor
13		Gears		Supplier	1	Steel	Motor
14		Vibrator Attachment Plate		In house	1	Steel	Vibration
15	Netter Vibration NEA5050	Vibrator		Supplier	1	Discrete Component	Vibration

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